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REVIEW ARTICLE

Repression of viral transcription during herpes simplex virus latency

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1. Introduction

After primary infection of an individual with herpes simplex virus (HSV) has subsided, the viral genome is retained in a nonreplicating state, known as latency, in sensory neurons. In response to the appropriate stimulus, the virus is reactivated to cause new cutaneous lesions. Many aspects of latency can be reproduced after infection of small animals such as mice, rabbits and guinea pigs. The topic has been a source of fascination for generations of virologists, but despite intense efforts over the past 30 years many questions remain. This review will focus on perhaps the most basic issue: how can a virus that appears ideally designed for lytic replication be retained in such a contrary state as latency? A subsidiary aim is to encourage a re-evaluation of the concept that tissue-culture studies are relevant to latency *in vivo*, a view that has never gained full acceptance by many those working on animal models of latency. First, some of the important aspects of HSV molecular biology and latency that relate to the theme of this review will be covered.

2. Lytic Infection

After infection with HSV, replication of virus and subsequent death of the host cell usually occur. The control of viral gene expression in this lifestyle is now well understood, and the salient points will be summarized here before commencing a discussion of latency. Although HSV encodes at least 74 unique genes (McGeoch *et al*., 1988; Dolan *et al*., 1998), only the five classified as immediate early (IE), plus that encoding the ribonucleotide reductase large subunit, are initially transcribed from the input genome. Four of the IE proteins participate in controlling gene expression, while one (named ICP47 or Vmw12) interferes with antigen presentation (York *et al*., 1994). The IE protein ICP4 (Vmw175) is essential for the transcription of early and late genes (Preston, 1979; Watson & Clements, 1980; DeLuca & Schaffer, 1985), acting through the basal cellular transcription machinery (Coen *et al*., 1986; Cook *et al*., 1995; Carrozza & DeLuca, 1996). ICP27 (Vmw63) is also required for virus replication, operating at post-transcriptional levels to regulate splicing, termination and nuclear export of viral transcripts (reviewed by Phelan & Clements, 1998). Although ICP0 (Vmw110) is not essential for virus replication, in its absence lytic infection is initiated inefficiently at low m.o.i., such that many genomes fail to give rise to progeny (Stow & Stow, 1986, 1989; Sacks & Schaffer, 1987; Everett, 1989; Cai *et al*., 1989). This protein stimulates the expression of all classes of viral genes (Cai & Schaffer, 1992; Chen & Silverstein, 1992) and, as discussed below, may play a key role in latency. The remaining IE species, ICP22 (Vmw68), is required for efficient replication of HSV in certain cultured cell types (Post & Roizman, 1981; Sears *et al*., 1985; Rice *et al*., 1995). Once IE proteins have been produced in adequate amounts, the infected cell is usually committed to transcription of the remainder of the viral genome, followed by DNA replication and synthesis of progeny virions. Without IE proteins, productive infection cannot occur.

Transcription of the IE genes is activated by VP16, a major component of the HSV tegument (reviewed by O'Hare, 1993). VP16 acts through the target sequence TAATGARAT (R is a purine nucleotide), which is present in at least one copy in all HSV IE promoters. TAATGARAT is a binding site for the cellular factor Oct-1, a member of a protein family initially characterized by the ability to bind the 'octamer' sequence ATGCAAAT (Sturm *et al*., 1988). It is now clear that proteins of the Oct family share a DNA-binding structure known as the POU domain, and that there is a great deal of flexibility in sequences recognized by the various Oct proteins (reviewed by Herr & Cleary, 1995; Ryan & Rosenberg, 1997). VP16 and an additional cell protein, HCF, bind to the Oct-1/TAATGARAT complex, thereby bringing the C-terminal domain of VP16, a strong transcription activating element, into proximity with the preinitiation complex. The reliance of VP16 on the host proteins Oct-1 and HCF for its effect on IE transcription suggests, as discussed later, that the availability of these cellular factors can influence the initiation of the HSV lytic cycle.

3. Latency

After infection of a human or experimental animal with HSV, virus replication occurs at the site of infection, presumably following the pattern of gene expression outlined

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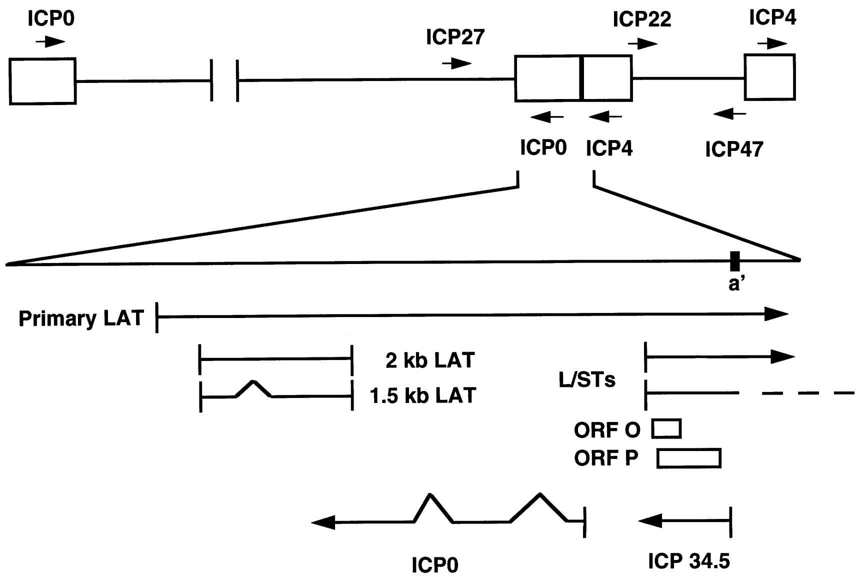

Fig. 1. Location of transcripts in the long repeat region of HSV-1 DNA. The internal repeat of HSV-1 is depicted, although the same gene arrangement exists in the terminal repeat.

above. Virus then enters nerve termini and is transported intra-axonally to the sensory ganglia, where infected neurons initially support virus replication. Within a few days, however, no free virus can be detected within ganglia and latency has been established (Stevens & Cook, 1971). The HSV genome is sequestered in this nonreplicating state in the neuronal nucleus, probably as a circular episome (Rock & Fraser, 1983, 1985; Efstathiou *et al*., 1986; Mellerick & Fraser, 1987), until signals which induce reactivation are received. Although the mechanism of reactivation is poorly understood, in general the known stimuli are linked by the ability to cause stress, either to the whole organism or directly to the neuron.

Analysis of viral gene expression during latency revealed that all lytic genes are switched off, but that one set of transcripts, known as the latency-associated transcripts (LATs), accumulates to high levels (Stevens *et al*., 1987; reviewed by Fraser *et al*., 1992; Feldman, 1994; Wagner *et al*., 1995; Wagner & Bloom, 1997). The major products are 2 kb and 1·5 kb RNAs, which are predominantly localized to the neuronal nucleus and are transcribed antisense to and partially complementary to the coding sequences for ICP0 (Fig. 1). These RNAs represent stable introns cleaved from a longer precursor that extends across the a′ sequence and into the short repeat region and is complementary to all of the ICP0 coding sequences (Farrell *et al*., 1991; Devi-Rao *et al*., 1991, 1994; Wu *et al*., 1996*b*; Rodahl & Haarr, 1997; Zabolotny *et al*., 1997; Arthur *et al*., 1998; Alvira *et al*., 1999). The reason for the continuing transcription of the LATs in an otherwise inactive genome is not known in detail, but it is clear that sequences located downstream of the promoter, and thus in the transcribed region, are important for this intriguing feature of the LATs region (Glorioso *et al*., 1992; Chen *et al*., 1995; Perng *et al*., 1996; Lachmann & Efstathiou, 1997; Lokensgard *et al*., 1997). There has been much attention to the structure of the LATs transcription unit and its relevance to latency, and this has been dealt with comprehensively in a recent review (Wagner & Bloom, 1997). Here, I will discuss only the aspects of LATs that bear on the repression of HSV gene expression during latency.

4. Measures of latency

Quantification of latency establishment is crucial to the interpretation of many of the experiments to be described in the following sections, yet this has been a very difficult area of research. One complexity resides in the indirect nature of the experimental systems currently available. Replication at the site of inoculation means that nerve termini are exposed to an essentially unquantifiable amount of virus over a period of days. Attempts to define the roles of individual genes in latency, for example by analysing the phenotypes of mutants in animal models, are complicated by the fact that most genes are important for optimal replication *in vivo*. A given virus mutant may exhibit an apparently lower efficiency of establishment of latency than wild-type virus when the real defect is in replication at the site of inoculation, resulting in a lower effective dose of virus applied to nerve termini. In addition, the various methods used to quantify latency highlight different aspects of the interaction.

Expression of LATs

The most obvious marker for latency is the presence of LATs, which unequivocally denotes the existence of latent HSV in a neuron. LATs can be quantified by analysing ganglion RNA on blots or by RT–PCR, and the proportion of neurons expressing LATs can be determined by *in situ* hybridization (ISH) (reviewed by Wagner & Bloom, 1997). There is good agreement between these different methods of estimating expression of LATs during latency.

Latent HSV DNA

Quantification of HSV DNA in ganglia can be achieved by solution hybridization (Puga *et al*., 1978), direct hybridization to viral probes on Southern blots (Rock & Fraser, 1983, 1985; Efstathiou *et al*., 1986; Simmons *et al*., 1992) or slot blots (Leib *et al*., 1989; Margolis *et al*., 1992), and by quantitative PCR (QPCR) (Katz *et al*., 1990; Sedarati *et al*., 1993; Ramakrishnan *et al*., 1994*a*, *b*). All of these approaches concur that infected neurons contain multiple copies of the viral genome since the number of molecules of HSV DNA detected exceeds the number of neurons in the ganglion, even though most neurons are not latently infected. The number of cells containing viral genomes has been estimated by application of *in situ* PCR to ganglion sections, and the data currently available show that more neurons contain viral DNA than score positive for LATs using conventional ISH, by a factor of between three and fifty (Gressens & Martin, 1994; Ramakrishnan *et al*., 1994*b*; Mehta *et al*., 1995; Maggioncalda *et al*., 1996). Many neurons which harbour HSV DNA are therefore phenotypically LATs negative (LAT^{-}), an observation that agrees with the initially surprising finding that LAT^{-} HSV mutants are able to establish latency (Wagner & Bloom, 1997). Evaluation of the situation is complicated by a report in which *in situ* RT–PCR was used to detect transcripts from the LATs region (Ramakrishnan *et al*., 1996). The number of neurons expressing LATs, as detected by this method, was greater than that found by conventional ISH and equivalent to the number positive for DNA by *in situ* PCR, suggesting that all infected neurons express LATs at some level. As will become apparent later, it is unfortunate that this study did not include analysis of some lytic cycle transcripts.

A further approach to analysing latent viral genome content is provided by the ingenious contextual analysis (CXA) technique (Sawtell, 1997). Mice are fixed by perfusion and ganglia dissociated into single cell suspensions. Neurons are purified by differential centrifugation and analysed by PCR, either as single cells or in batches, thereby enabling the proportion of DNA-containing cells, and HSV genome copy-number in individual neurons, to be calculated. This technique confirms that many more neurons contain viral DNA than are positive for LATs by ISH, and additionally shows that the copy-number in individual neurons varies from one to thousands. This wide variation is problematic because it implies that other methods of quantifying viral DNA pertain only to the small population of neurons containing the very high copy-numbers. It will be important to extend the CXA approach to determine how expression of LATs is related to genome copy-number in individual neurons.

Reactivation

Titration of virus produced upon induction of reactivation was the first method used to quantify latency, and it could, arguably, still be considered the most relevant approach. The most widely used experimental system to provoke reactivation is explantation of ganglia and subsequent culture in the laboratory (Stevens & Cook, 1971), after which the proportion of ganglia that release HSV, and the time of culture required before virus is detected, are determined. The explantation procedure invariably provokes reactivation of replication-competent HSV, but it is such a harsh treatment that the relevance to events in the animal or human can be questioned. Periodic spontaneous shedding of virus occurs after infection of rabbits or guinea pigs, but not mice, and reactivation *in vivo* can be induced in mice and other animals by a number of stressful treatments (Walz *et al*., 1974; Openshaw *et al*., 1979; Kwon *et al*., 1981; Stanberry *et al*., 1982; Shimeld *et al*., 1990; Sawtell & Thompson, 1992*b*). The use of reactivation to quantify latency is troublesome. A major problem is the general inefficiency of the process, with only a few cells responding in ganglia known to harbour the HSV genome in up to 25% of neurons (McLennan & Darby, 1980; Sawtell & Thompson, 1992*b*; Ecob-Prince *et al*., 1993*b*; Kosz-Vnenchak *et al*., 1993; Ecob-Prince & Hassan, 1994; Sawtell, 1997). This finding complicates the interpretation of many experiments because the possibility always exists that only a subset of the genomes detected during latency is competent for reactivation while other copies, maybe the majority, represent a 'dead end' sequestration of viral DNA. In studies of reactivation in the mouse *in vivo* and after explantation, ICP0-specific RNA was found predominantly in LAT^{-} neurons, suggesting that reactivation competence does not correlate with expression of LATs in individual neurons (Ecob-Prince *et al*., 1993*b*; Ecob-Prince & Hassan, 1994). The efficiency of reactivation does, however, correlate with the amount of viral DNA in ganglia, both in terms of the number of neurons containing HSV genomes and the average copy-number per positive cell (Lekstrom-Himes *et al*., 1998; Sawtell, 1998; Sawtell *et al*., 1998).

A further complication is that the assays used to date measure gene expression or virus replication which occurs as a consequence of reactivation, but do not necessarily focus on the primary changes to the viral genome that switch it out of latency. For example, there is agreement that HSV mutants lacking functional ICP0 reactivate poorly (Leib *et al*., 1989; Clements & Stow, 1989; Gordon *et al*., 1990; Cai *et al*., 1993), but there are no data to distinguish whether the role of ICP0 is in converting the latent genome from a silent to an active state or in assisting gene expression in the first and subsequent rounds of replication *after the primary changes have occurred*. Examination of expression from genomes unable to synthesize viral DNA during reactivation showed only low levels of all classes of viral transcripts, indicating that analysis of the earliest events will be difficult (Kosz-Vnenchak *et al*., 1993). Nonetheless, there is a need for the development of systems to identify and quantify the primary changes to viral gene expression that occur upon induction of reactivation.

In summary, neurons harbour multiple copies of the HSV genome, with a wide range of copy-numbers in individual cells. Not all latently infected neurons express LATs (as detected by ISH), thus only a proportion of viral genomes have the LATs transcription unit detectably active during latency. Reactivation can be provoked in only a small fraction of the neurons that contain viral DNA. Reactivation does not absolutely depend on the presence of LATs. All of these factors should be taken into account when attempting to quantify the establishment of latency by HSV.

5. A block at the IE level

Given the facility with which HSV infects and kills cells in culture, it is surprising that a lifestyle as contrasting as latency can exist. Neurons are also susceptible to lytic infection by HSV *in vivo* yet in the long term this is the cell type in which latent genomes are retained. Thus the mechanism by which transcription of the genome is prevented in neurons during latency is of central importance. It is thought that the outcome of infection is determined at an early stage in the interaction of HSV with neurons, and that lytic infection and latency represent separate pathways. During the first few days after footpad infection of mice, neurons of the dorsal root ganglia (DRG) can be defined as expressing either productive cycle gene products or transcripts from the LATs promoter (Margolis *et al*., 1992). Additionally, after flank inoculation of mice both LATs and lytic cycle gene products can be detected in ganglia which innervate the site of inoculation, but only LATs are found in peripheral neurons not connected to the infected dermatome (Speck & Simmons, 1991, 1992). Similar findings were obtained using the mouse-ear inoculation model (Lachmann *et al*., 1999). Together, these studies suggest that latency is established without detectable lytic gene expression and hence that the commitment to latency occurs shortly after infection of the neuron. This concept is reinforced by the observation that latency ensues after infection of mice with virus mutants that lack functional ICP4 and are thus unable to express early and late proteins (Dobson *et al*., 1990; Katz *et al*., 1990; Sedarati *et al*., 1993). The hypothesis that latency and lytic infection are alternative, mutually exclusive, outcomes after infection of neurons with HSV underlies most current thinking on latency.

It is suspected that the block to virus replication occurs at the level of expression or function of IE gene products, and two lines of evidence have led to the idea that a failure of IE transcription may be the key. One is that in tissue-culture cells expression of HSV IE proteins is invariably cytotoxic, a property which, if it applies to neurons, is incompatible with latency (Johnson *et al*., 1992, 1994; Wu *et al*., 1996*a*; Preston *et al*., 1997; Samaniego *et al*., 1997, 1998). A second line of evidence is provided by use of *in*1814, an HSV-1 mutant with a mutation that abolishes the interaction of VP16 with Oct-1 and HCF, resulting in reduced IE gene expression (Ace *et al*., 1989). In tissue-culture cells, *in*1814 can replicate but initiates infection inefficiently at low m.o.i., like ICP0-deficient mutants, showing that IE transcription is not absolutely dependent on VP16. When inoculated into mice, *in*1814 replicates poorly at the site of inoculation and in neurons of the sensory ganglia, yet latency is established in at least as many cells as found with wild-type HSV-1 (Steiner *et al*., 1990; Valyi-Nagy *et al*., 1991*b*; 1992; Ecob-Prince *et al*., 1993*a*). Although many factors, such as differential peripheral replication, preclude an accurate quantitative statement on the efficiency of establishment of latency by *in*1814, an obvious message from these experiments is that transactivation of IE genes by VP16 is not absolutely required for establishment of latency, in agreement with the hypothesis that the natural block is at the IE level. I will now discuss ideas relating to the mechanisms by which IE gene expression may be restricted in neurons.

VP16 function in neurons

The finding that *in*1814 can establish latency is compatible with the concept that VP16 fails to transactivate IE transcription in wild-type HSV-1-infected neurons, resulting in the initiation of lytic infection in only some neurons and establishment of latency in others. This is similar to the fate of *in*1814-infected tissue-culture cells, in which only a proportion (depending on the initial m.o.i.) of cells support virus replication. Failure to transactivate IE transcription in neurons could result from the absence of VP16 function, or from a problem with the cell factors Oct-1 and HCF. The fact that VP16 is a tegument protein, and that it can act in *trans*, led to the proposal that it is not transported with the viral nucleocapsid during the long journey along the axon to the neuronal nucleus, and thus that activation of IE transcription would not occur (Roizman & Sears, 1987; Kristie & Roizman, 1988). To test this hypothesis the VP16 open reading frame was placed under the control of the metallothionein (MT) promoter, or cloned in the antisense orientation, and the constructs were inserted into the HSV-1 genome (Sears *et al*., 1991). Mice were infected with the HSV-1 recombinants and fed cadmium salts to induce VP16 synthesis. There was no difference in the efficiency of latency establishment, irrespective of cadmium feeding, by the two viruses. In addition, mice transgenic for the MT promoter–VP16 construct were produced and infected, but again latency was not influenced by feeding the animals cadmium during the early stages of infection. The authors concluded that latency could be established even in the presence of functional VP16, and therefore that latency does not result solely from the absence of this protein. Two considerations suggest caution in interpreting these experiments. First, it was claimed that the MT promoter directed long-term expression of β-galactosidase (when cloned upstream of the *Escherichia coli lacZ* coding sequences) and of VP16-specific RNA during latency, a startling observation in view of the typical finding that promoters cloned into the HSV-1 genome are inactive during latency. In the case of β-galactosidase controlled by the MT promoter, this claim was not confirmed by others (Lokensgard *et al*., 1994). Second, no evidence was presented by Sears *et al*. (1991) for the presence of VP16 protein in neurons that received viral genomes. A reinvestigation of the ability of VP16 to function in neurons *in vivo* is warranted, and there is also a need for a detailed investigation of the fate of incoming VP16 in infected mice to be carried out.

The Oct factor profile of sensory neurons

ISH studies suggested that Oct-1 was present only at low abundance in sensory neurons and that other proteins of the Oct family were expressed preferentially (He *et al*., 1989). Since many members of the Oct family can bind to TAATGARAT elements but only Oct-1 is known to interact with VP16, other Oct proteins might act as competitors to prevent formation of the VP16/Oct-1/HCF complex and thus repress IE transcription. The validity of this idea was supported by studies on HSV-1 infection of the C1300 mouse neuroblastoma cell line and of hybrid cell lines (ND cells) made by fusing C1300 with rat DRG neurons. The C1300 and ND lines contain neuron-specific forms of Oct-2 (Oct-2.4 and Oct-2.5) which bind to TAATGARAT elements but do not associate with VP16, thereby acting as repressors of IE-specific gene expression in transfection assays (Kemp *et al*., 1990; Dent *et al*., 1991; Lillycrop *et al*., 1991, 1993; Lillycrop & Latchman, 1992). It was proposed that the presence of the Oct-2 isoforms accounts for the relative nonpermissiveness of C1300 and ND cells to lytic infection by HSV-1 (Kemp & Latchman, 1989; Kemp *et al*., 1990; Wheatley *et al*., 1990; Lillycrop *et al*., 1991, 1994). Oct-2 was detected in cultured adult rat DRG neurons by electrophoretic mobility shift assay (EMSA), leading to the suggestion that neuronal forms of this protein repress transcription of HSV IE genes *in vivo*, resulting in latency (Wood *et al*., 1992). In contrast, Hagmann *et al*. (1995) detected Oct-1 (at low levels) but no Oct-2 in extracts of newborn mouse DRG neurons and in extracts of adult pig sensory ganglia, by EMSA. These authors were also unable to reproduce the repression of IE gene expression by Oct-2.5 in transfection assays. To add to the inconsistencies, Turner *et al*. (1996) could detect, with a ribonuclease protection assay, Oct-1- but not Oct-2-specific RNA in trigeminal ganglia from adult mice, but Wood *et al*. (1992) were able to amplify Oct-2-specific sequences from adult rat DRG neurons by RT–PCR. A clear consensus on the importance, or otherwise, of Oct-2 isoforms for repression of HSV IE gene expression is needed to resolve the status of this potentially important issue. Apart from Oct-2, there are many other POU domain proteins in the peripheral and central nervous systems (reviewed by Latchman, 1999), some of which would be expected to bind TAATGARAT and possibly to alter IE transcription. It should be noted, however, that competition with Oct-1 for binding to TAATGARAT will only affect IE gene expression if VP16 is functional, for if VP16 does not operate in neurons then other Oct proteins will exert significant effects only if they repress basal transcription from IE promoters.

A further consideration is that mouse Oct-1 differs from human Oct-1 in crucial amino acids that mediate the interaction with VP16, such that IE promoters are less responsive to VP16 in mouse cells than in human cells (Cleary *et al*., 1993; Suzuki *et al*., 1993). If activation of IE transcription by VP16 can occur in neurons, it might be expected that productive replication would be relatively favoured in the natural host.

In an intriguing set of experiments, Valyi-Nagy *et al*. (1991*a*) showed that RNAs specific for the cellular factors c-Fos, c-Jun and Oct-1 were induced in a proportion of trigeminal ganglion neurons very shortly after corneal inoculation of HSV-1, and that c-Jun- and Oct-1-specific RNAs were produced even when corneal scarification was carried out without virus. The increased Oct-1 levels, induced in neurons by the peripheral stimulus, might tip the balance between lytic and latent infection when the virus arrives at the ganglion, by competing with repressive Oct family proteins or, if VP16 is functional in neurons, by stimulating IE gene expression. Alternatively, the signals which induce expression of c-Jun and c-Fos may activate HSV IE transcription and favour lytic replication, as occurs in tissue-culture cells (Preston *et al*., 1998).

HCF

Neuron-specific differences in the activity of HCF could account for low IE transcription. The profile of the TAATGARAT-specific complexes formed in EMSA with DRG extracts differed from that with HeLa cell extracts, and it was proposed that this finding may signify the existence of altered forms of HCF which may be important functionally in IE gene expression (Hagmann *et al*., 1995). More recent studies have shown that HCF is sequestered exclusively in the cytoplasm of sensory neurons (Kristie *et al*., 1999), as opposed to a general or nuclear distribution in other cell types (Kristie *et al*., 1995; La Boissière *et al*., 1999). Indeed, HCF has an important role in transporting VP16 to the cell nucleus (La Boissière *et al*., 1999) and it is thus possible that the neuronal forms of HCF may fail to take VP16 to the nucleus. The consequence of this failure could be inefficient transcription of IE genes.

Repression of IE gene expression by LATs

Ever since the discovery of LATs, attempts have been made to link the transcripts with some aspect of latency, largely by studying the properties of LAT$^-$ virus mutants (reviewed by Fraser *et al*., 1992; Wagner & Bloom, 1997). This field is complex and largely beyond the remit of this review, but it is a fair generalization to state that LAT$^-$ virus mutants are impaired for reactivation *in vivo*. In explantation models, the data contain many apparent conflicts that are difficult to reconcile, with some LAT$^-$ mutants exhibiting impaired reactivation efficiency and others behaving normally. The crucial question for discussion here is whether, in those examples in which the absence of LATs correlates with a reduced efficiency of reactivation, the phenotype is due to an effect on reactivation per se, or whether it is an indirect consequence of inefficient latency establishment by LAT$^-$ mutants, thereby loading ganglia with fewer viral genomes.

As soon as the genetic organization of LATs had been unravelled it was suspected that the transcripts could inhibit the production of ICP0 by an antisense mechanism, and this view was supported by the finding that expression of the 2 kb species of LATs inhibited, in *trans*, the activation of the HSV-1 thymidine kinase (TK) promoter by ICP0 in transfected cells (Farrell *et al*., 1991). It followed that, in the absence of LATs, more ICP0 might be produced and hence a greater proportion of neurons may become lytically infected and subsequently eliminated, leaving a smaller pool of latent viral genomes. This view was supported by experiments which showed that mice infected with LAT$^-$ mutants contained more productively infected neurons than those infected with parental LATs positive (LAT$^+$) viruses (Sawtell & Thompson, 1992*a*). Interestingly, the difference was observed after corneal, but not footpad, infection, suggesting that there is variation in the responses of neurons at different anatomical locations. Garber *et al*. (1997) found that IE and early RNAs accumulated more rapidly in trigeminal ganglion neurons after infection with LAT$^-$ mutants than with rescued revertants and that approximately 1·5 times more neurons expressed these transcripts at early times after infection with the mutant, in support of the view that LATs have a role in suppressing lytic gene expression. The active component of the LATs transcription unit is unlikely to be solely the major 2 kb species, based on a report that a mutant lacking most of this transcript establishes latency and reactivates normally (Maggioncalda *et al*., 1996).

A crucial prediction of the hypothesis that lack of LATs derepresses IE gene expression and hence permits a greater level of lytic infection is that fewer reactivatable viral genomes should remain after infection with LAT$^-$ mutants. Using the same experimental system as Garber *et al*. (1997), Chen *et al*. (1997) found that the amounts of viral DNA retained during latency with a poorly reactivating LAT$^-$ mutant and rescued revertant were indistinguishable by QPCR. Similarly, QPCR detected a reduction of no more than 30% in retention of HSV DNA by a mutant impaired for reactivation (Devi-Rao *et al*., 1994). With a different strain of HSV-1, Thompson & Sawtell (1997) used CXA and found that the number of neurons retaining viral genomes was reduced by approximately 3-fold in ganglia of LAT$^-$ virus-infected mice, compared with those from animals inoculated with a rescued revertant. The copy-number profile of the infected neurons, measured by CXA, was unaltered, essentially meaning that the absence of LATs reduced the pool of HSV-containing neurons without changing the mean number of viral genomes per infected neuron. Clearly, these results support the view that LATs improve the establishment of latency by suppressing the synthesis of ICP0 and hence of other lytic cycle gene products, but equally clearly they conflict with the findings of Chen *et al*. (1997) and Devi-Rao *et al*. (1994), which claim that the amounts of DNA retained during latency by wild-type HSV-1 and LAT$^-$ mutants are essentially indistinguishable. It may simply be that the techniques used to quantify viral DNA have inherently different limitations especially since, in a separate study, CXA revealed differences in DNA levels which, in parallel ganglion samples, were not detected by QPCR (Sawtell *et al*., 1998). Whichever of the views on genome retention after infection with LAT$^-$ mutants holds, there are further difficult questions to answer when other data are considered. The results from *in situ* PCR and CXA suggest that 67–95% of neurons containing HSV DNA do not express LATs that are detectable by ISH (Gressens & Martin, 1994; Ramakrishnan *et al*., 1994; Mehta *et al*., 1995; Maggioncalda *et al*., 1996; Sawtell, 1997). If the absence of LATs results in a 3-fold reduction in the number of genome-containing neurons due to greater virus replication and cell death, this must occur even in neurons that are destined not to express LATs (at levels detectable by ISH). On the other hand, if LATs suppress lytic functions but there is no difference in the retention of viral DNA after infection with LAT$^-$ compared with LAT$^+$ viruses, it follows that only a specific subpopulation of neurons, taken largely from those most likely to reactivate, is destroyed by the more vigorous replication of LAT$^-$ mutants.

The idea that a product of the LATs region represses IE gene expression is supported by analysis of neuronally derived cell lines expressing the LATs transcription unit (Mador *et al*., 1998). These lines constitutively produce LATs, especially the major 2 kb species, and are relatively nonpermissive for replication of HSV-1 after infection at low m.o.i. Furthermore, accumulation of RNAs encoding ICP0, ICP4 and ICP27 is greatly reduced compared with that found after infection of a control transformed cell line, although it is not clear whether the effect on ICP4- and ICP27-specific RNAs is directly related to the presence of LATs or is a consequence of a reduction in ICP0 protein levels.

The role of LATs in HSV latency is still a complex issue. Despite the attractions of the models described above, it remains an uncomfortable truth that LATs are not absolutely required for any aspect of latency that has been monitored to date. It is possible that the true functions of these RNAs cannot be reproduced in animal models.

ORFs O and P

A set of virus-specified RNAs, named L/STs, has been described by two groups (Bohensky *et al*., 1993, 1995; Yeh & Schaffer, 1993). These RNAs are transcribed divergently from ICP0 mRNA and overlap, in the sense orientation, the LATs primary transcript (Fig. 1). The L/STs exhibit a novel pattern of regulation since they are not detected during infection in the absence of protein synthesis but are overproduced during infection with viruses in which ICP4 is inactive. In parallel with the transcript analysis, it was found that an open reading frame (named ORF P) encompassed by the L/STs was expressed only during infection with HSV-1 ICP4 mutants (Lagunoff & Roizman, 1994). Intriguingly, ORF P almost exactly overlaps (in the antisense orientation) the coding sequences for the neurovirulence factor ICP34.5. A strong binding site for ICP4 is found at the 5′ end of the L/STs (Yeh & Schaffer, 1993; Samaniego *et al*., 1995; Randall *et al*., 1997), and mutagenesis of this site resulted in overproduction of ORF P and L/STs during normal infection, demonstrating that ICP4 represses transcription of L/STs (Lagunoff & Roizman, 1995; Lee & Schaffer, 1998). More recently, a less abundant product, ORF O, has been detected (Randall *et al*., 1997). This protein is encoded by sequences which overlap, in the same orientation but a different translation frame, the coding region for ORF P. Synthesis of the mRNAs encoding ORFs O and P appears to be controlled by the same promoter.

The unusual regulation of the L/STs led Lagunoff & Roizman (1995) to point out that expression of ORFs O and P would only be expected in infected cells lacking functional ICP4, a condition satisfied by latency. Further studies showed that the ORF P product binds to the splicing factor SF2/ASF and thus might inhibit the accumulation of the spliced mRNAs encoding ICP0 and ICP22 early in infection (Bruni & Roizman, 1996). The evidence relating to this proposal is controversial, relying on the analysis of IE mRNA and protein accumulation in cells infected with HSV-1 mutants that have the L/STs ICP4 binding site mutated, causing overproduction of the ORF P product. When expression of ORF P was derepressed in this way, Bruni & Roizman (1996) found a transient reduction in ICP0 and ICP22 protein levels, but Lee & Schaffer (1998) did not detect any differences in the accumulation of cytoplasmic ICP0-specific RNA. A fusion protein containing the ORF O sequences linked to glutathione *S*-transferase interacted with ICP4 and, at high concentrations, prevented it from binding to the L/STs repressor site *in vitro* (Randall *et al*., 1997). Together, the observations predict that a positive feedback system could control HSV gene expression during latency: in the absence of ICP4, ORFs O and P would be produced and would prevent the synthesis or function of IE proteins, thereby maintaining the remainder of the genome in a stable nontranscribed state. Conversely, during productive infection ICP4 represses production of L/STs. Hypothetically, reactivation could result from interference with ORF O or P function during latency. While this is an elegant model, some caveats must be registered. First, latency was established apparently normally by a virus mutant in which the ORF P reading frame was specifically interrupted (Lee & Schaffer, 1998); second, mutants with derepressed synthesis of ORF P established latency inefficiently but they were also highly attenuated, suggesting that the consequence of overexpressing ORF P is reduced replication at the periphery or in the ganglia (Lagunoff *et al*., 1996; Lee & Schaffer, 1998); and third, there is as yet no evidence for expression of ORFs O or P in neurons harbouring latent HSV genomes.

As with LATs, the ideas discussed in this section must take heed of the fact that latency can be established, apparently normally, in mice by HSV-1 mutants in which ORFs O and P have been deleted.

6. Consequences of an IE block

While there is a reasonable degree of acceptance that inefficient IE gene expression or function is a prerequisite for the establishment of latency, there are conceptual differences of opinion concerning the functional state of the latent genome, and in particular the accessibility of viral promoters to activating factors. One view, following largely from tissue-culture studies, envisages a global repression of the HSV genome which requires major changes to reverse it. The other, based on studies from animal models and cultured neurons, considers the programme of gene expression essentially to be stalled, with the genome, or at least strategic promoters, remaining responsive to signals provided by neurons.

Global repression in the absence of IE proteins

The availability of *in*1814 provides a model to study the consequences of blocking IE transcription (Ace *et al*., 1989). More recently, additional mutations that inactivate ICP4 and ICP0 function have been introduced into the *in*1814 genome and, in separate studies, deletion mutants lacking the genes encoding ICP4, ICP0 and ICP27 have been constructed (Jamieson *et al*., 1995; Preston *et al*., 1997; Samaniego *et al*., 1997, 1998). When tissue-culture cells are infected with mutants of this type, lytic gene expression is not observed and the viral genome is sequestered as a nonlinear molecule, as found during latency *in vivo*, for many days (Harris & Preston, 1991; Jamieson *et al*., 1995; Preston & Nicholl, 1997; Samaniego *et al*., 1998). No viral gene expression, including that of LATs, is detectable in this quiescent state and functional changes in the HSV genome can be detected. Immediately after infection with mutants impaired for IE gene expression, promoters in the viral genome are, as expected, responsive to transacting factors: the HSV IE promoters can be switched on by provision of VP16, and the human cytomegalovirus (HCMV) IE promoter (cloned into the HSV genome) is activated by one or more proteins present in the HCMV virion. If, however, the same transactivating proteins are provided 24 h after infection of cells with the IE-deficient mutant, the promoters are no longer responsive (Harris & Preston, 1991; Preston & Nicholl, 1997; Samaniego *et al*., 1998). In the absence of functional IE proteins, the HSV genome is converted to a quiescent, repressed state in which gene expression is switched off and is no longer responsive to control mechanisms which operate if applied at the time of infection. Indeed, the quiescent genome is remarkably resistant to reactivation by a variety of stimuli, including division of the host cell, emphasizing the stability of the sequestration (Jamieson *et al*., 1995; C. M. Preston, unpublished observations).

Two lines of evidence suggest that repression is not achieved through DNA sequence-specific interactions. First, the four HSV IE promoters and the HCMV IE promoter are shut off, even though there are no sequence motifs obviously common among them apart from the TATA box (Preston & Nicholl, 1997; Samaniego *et al*., 1998); second, strong cellular promoters, such as that controlling the expression of β-actin, are inactive when inserted into an IE-defective HSV mutant (C. M. Preston, unpublished observations). The latter observation suggests that there are differences in the activities of promoters depending on their environments (HSV or cell genomes), and this proposal is supported by the finding that the HSV-1 ICP0 promoter is repressed when cloned into the HSV-1 genome but not when present in the cell genome driving expression of a selected marker (Preston & Nicholl, 1997).

Similarly, *in vivo*, it has been shown in numerous studies that neuron-specific promoters and strong cell promoters are repressed during latency when cloned into the HSV genome, even though the copies in the host genome presumably remain active (Glorioso *et al*., 1995; Lachmann *et al*., 1996). The structural basis for the repression is not known at present: it is generally assumed that the viral genome, in the absence of IE proteins, is treated by the cell as a set of genes destined to be silenced. The quiescent state observed in tissue-culture systems has obvious similarities to latency *in vivo* and it may be reasonable to conclude that a failure to express IE proteins inevitably leads to repression of the HSV genome in neurons, resulting in latency.

A key element in understanding the attainment of the quiescent state concerns the mode of action of ICP0. Prior to the construction of *in*1814, deletion mutants lacking ICP0 coding sequences were isolated and found to exhibit a propensity to reside in cells in a nonreplicating form after infection at low m.o.i. (Stow & Stow, 1986, 1989; Sacks & Schaffer, 1987; Everett, 1989). The mutant genomes could remain for many days and were recovered by superinfection of cultures with HCMV or varicella-zoster virus, demonstrating that they were not permanently inactivated and that absence of ICP0 function is sufficient for the genome to become quiescent (Stow & Stow, 1989). The quiescent state attained by IE-deficient mutants could be reversed by provision of ICP0, and indeed this protein (and its functional homologues in other herpesviruses) is the only agent currently known that can overcome the repression of gene expression and allow replication to resume (Stow & Stow, 1989; Harris & Preston, 1991; Preston & Nicholl, 1997; Everett *et al*., 1998*b*; Samaniego *et al*., 1998). This observation points to a key role for ICP0 in determining the outcome of infection in tissue-culture cells: in its absence, conversion to the quiescent state occurs, while in the presence of ICP0 repression is prevented and, indeed, reversed such that lytic replication ensues. It follows that a detailed description of the properties of ICP0 should lead to an understanding of the mechanism of repression.

ICP0 was initially classified as a nonspecific transcription activator (Everett, 1984; O'Hare & Hayward, 1985), but more recent studies suggest that the protein acts to alter the intranuclear environment to favour transcription of the HSV genome. The presence of functional ICP0 in cells causes the rapid disruption of nuclear substructures known as nuclear domain 10 [ND10; also named promyelocytic leukaemia (PML) bodies or PML oncogenic domains (PODs)] (Maul *et al*., 1993; Maul & Everett, 1994; Everett & Maul, 1994). This initially curious result assumed relevance when it was shown that input DNA of HSV (and other viruses) is rapidly transported to ND10 after infection, linking the disruption of the substructures with potential effects on the function of the viral genome (Maul *et al*., 1996; Maul, 1998). It was further shown that ICP0 induces the degradation of isoforms of the cellular protein PML (a major constituent of ND10) that are conjugated with the small ubiquitin-like protein SUMO-1, and that this degradation is blocked by MG132, an inhibitor of the activity of the 26S proteasome (Everett *et al*., 1998*a*; Chelbi-Alix & de The, 1999; Muller & Dejean, 1999). Crucially, when MG132 was included in the culture medium the pattern of infection with wild-type HSV-1 was altered to resemble that of an ICP0 null mutant: early gene expression was inhibited, ND10 were not disrupted, and the reactivation of quiescent genomes was abrogated (Everett *et al*., 1998*a*, *b*). The activity of the 26S proteasome is therefore critical for ICP0 function, leading to the hypothesis that ICP0 acts by inducing the degradation of one or more of the proteins involved, either directly or indirectly, in genome repression.

Clues to the identities of the important targets for ICP0 have recently emerged, arising from the work of R. D. Everett and colleagues. The kinetochore/centromere protein CENP-C is also destroyed in response to ICP0, resulting in the disruption of mitosis (Everett *et al*., 1999*a*). The centromere is a heterochromatic site, and there is evidence that ND10 can have a short-lived association with centromeres during the G_2 phase of the cell cycle (Everett *et al*., 1999*b*). In addition, Sp100, another SUMO-1-modified ND10 component which is degraded in response to ICP0 (Chelbi-Alix & de The, 1999; Muller & Dejean, 1999), forms a complex with the heterochromatin-associated protein HP-1 (Lehming *et al*., 1998; Seeler *et al*., 1998). Protein HP-1, in *Drosophila*, promotes position-effect variegation, a phenomenon in which large arrays of genes are silenced (James *et al*., 1989; Eissenberg *et al*., 1992). It is therefore possible to trace links between the intranuclear effects of ICP0 and possible disruption of cellular mechanisms for silencing genes, and thus to speculate that ICP0, perhaps through its localization to ND10, disturbs the interaction of HP-1 or other cellular proteins with heterochromatin. Mechanisms along these lines must take account of the observation that the β-globin gene is thought to be packaged in a tight chromatin structure in nonerythroid cells but it is not transcribed after infection with HSV-1, even though the promoter is responsive to IE proteins (Everett, 1985; Cheung *et al.*, 1997). It is not known whether the HSV genome is organized into a structure similar to heterochromatin during latency: the only reports available suggest a regular nucleosomal organization in brain stem *in vivo* (Deshmane & Fraser, 1989) but not during quiescent retention in fibroblasts (Jamieson *et al.*, 1995). If a heterochromatin-like structure accounts for genome silencing during latency, the LATs region must possess interesting properties to overcome it at a local level.

The recent advances in understanding the mode of action of ICP0 support the view that major changes in the gross organization of the quiescent genome are required for the resumption of viral gene expression.

Since VP16 and ICP0 are not detectably expressed during latency in neurons, the initial events in reactivation must occur in the absence of these proteins. Studies by P. A. Schaffer's group showed that tissue-culture cells are more permissive for the replication of ICP0 null mutants, and transcribe viral IE genes more efficiently, in G_1 phase a few hours after release from growth arrest (Cai & Schaffer, 1991; Ralph *et al.*, 1994). In addition, ICP0 is not required in certain cell types, for example the osteosarcoma line U2OS, since viral null mutants initiate infection as efficiently as wild-type HSV-1 (Yao & Schaffer, 1995). In the rat pheochromocytoma cell line PC12, activation of signal transduction pathways by treatment with nerve growth factor (NGF) or fibroblast growth factor increased permissiveness to ICP0 null mutants (Jordan *et al.*, 1998). These studies suggest that certain host proteins are synthesized or rendered functional in response to cellular signals, and that they act analogously to ICP0, perhaps via proteolysis. Reactivation from latency may be mediated by cell proteins of this type.

In most tissue-culture cells, a block to IE gene expression results in attainment of the quiescent state in which the entire genome is repressed. This state is overtly equivalent to that reached during latency *in vivo*, with the exception that the LATs transcription unit is not active. Strict extrapolation from the tissue-culture studies leads to the prediction that promoters within the latent genome *in vivo* should be similarly unresponsive to *trans*-acting factors, and that reactivation occurs through a major structural reorganization of the genome. Such reorganization could occur by modification or synthesis of cellular functional homologues of ICP0, by degradation of the crucial targets for ICP0, or possibly by specific activation of ICP0 synthesis from the latent genome.

Responsive genomes during latency

An alternative vision of latency considers the HSV genome to be untranscribed, due to inadequate amounts of IE proteins, but inherently responsive to the appropriate cellular signalling pathways. This view is supported by the work of C. L. Wilcox and colleagues, who have used cultured foetal DRG neurons from rats, or other mammals including humans, to develop an *in vitro* system from the natural host cell for HSV latency. The neurons are inherently susceptible to infection with HSV-1, although at low m.o.i. some are not killed (Wilcox & Johnson, 1987). At m.o.i.'s up to 5 p.f.u. per cell, virus replication and cytotoxicity can be overcome by treating cultures with an inhibitor of DNA synthesis (usually acycloguanosine) for the first 7 days after infection, and no virus replication is detected even for 5 weeks after removal of acycloguanosine, provided NGF is present in the culture medium (Wilcox & Johnson, 1987, 1988). Reassuringly, most neurons express the 2 kb species of LATs (Smith *et al.*, 1994). Reactivation can be induced readily by the withdrawal of NGF for times as short as 1 h, or by treatment with activators of protein kinases even in the presence of NGF (Wilcox & Johnson, 1987, 1988; Wilcox *et al.*, 1990; Smith *et al.*, 1992). Both of these treatments affect neuronal gene expression and thus may also activate the viral genome. Somewhat surprisingly, expression of ICP0 strongly enhances the establishment of latency in cultured neurons. HSV-1 ICP0-defective mutants established latency approximately 1000-fold less efficiently (on the basis of input genome numbers) than a wild-type virus, and this defect was overcome by coinfection with an adenovirus recombinant that expresses ICP0 (Wilcox *et al.*, 1997).

A similar interaction to that observed after infection of cultured neurons can be obtained by incubating dissociated ganglia from latently infected mice in the presence of an inhibitor of DNA synthesis (Moriya *et al.*, 1994; Halford *et al.*, 1996). Upon removal of the inhibitor, HSV remains latent but can be reactivated by treatments that disturb cell physiology, for example heat shock or addition of dexamethasone to cultures. Interestingly, elevating cAMP levels, which reactivates HSV in the Wilcox system, was without effect in dissociated ganglia.

Experiments carried out on ganglia from latently infected mice suggest that an unusual regulatory pathway exists in neurons. Initially, it was shown that TK-negative (TK^-) mutants were able to replicate in the cornea but not in trigeminal ganglion neurons, presumably due to the absence of the precursors for DNA replication in the neuron, a non-dividing cell (Field & Wildy, 1978; Coen *et al.*, 1989; Efstathiou *et al.*, 1989; Leist *et al.*, 1989; Tenser *et al.*, 1989). This finding provided an experimental system to investigate the influence of viral DNA synthesis, in the neuron, on latency. Analysis of gene expression after infection with TK^- mutants revealed, surprisingly, that only low amounts of viral transcripts in all kinetic classes were detectable by ISH in neurons at 3–4 days after inoculation, compared with the situation after infection with wild-type HSV-1 (Kosz-Vnenchak *et al.*, 1990). Inhibition of viral TK or DNA polymerase during explant reactivation of HSV-1 also resulted in reduced IE and early transcript accumulation (Kosz-Vnenchak *et al.*, 1993). In contrast, when tissue-culture cells are infected and viral DNA synthesis is inhibited, late protein synthesis is reduced but IE and early transcript levels are at least equal to those in untreated cells (Honess & Roizman, 1974). These observations prompted the hypothesis that HSV gene regulation is different in neurons. Starting with the assumption that IE transcription is inefficient, Kosz-Vnenchak *et al*. (1990) proposed that this leads to a reduction in early gene expression, producing amounts of early proteins that are too low to initiate viral DNA synthesis. Viral genome replication per se is proposed to be a trigger which leads to full IE, and hence early, gene expression, with failure to initiate DNA synthesis the crucial event preventing HSV from undergoing productive infection. Reactivation could occur through stimulation of IE gene expression, but the fact that IE transcripts are not produced in detectable (by ISH) amounts after explantation reactivation when DNA synthesis is inhibited suggested an alternative route (Kosz-Vnenchak *et al*., 1993). It may be that early gene expression is directly activated, resulting in viral DNA synthesis and consequent raising of IE protein levels. The existence of this novel pathway of gene regulation in neurons was supported and extended by the application of RT–PCR during the early stages of infection of mice (Kramer *et al*., 1998). The number of IE (ICP4) and early (TK) transcripts per ganglion was approximately equal in mice inoculated with either wild-type HSV-1 (KOS) or TK^- mutants, and this number increased gradually throughout the first 32 h after infection for each virus. These transcripts were thought to arise mainly from input genomes, since the HSV-1 DNA levels were comparable for the two viruses during these early stages of infection, and they presumably represent the low levels of IE and early species that were not detectable by Kosz-Vnenchak *et al*. (1990) using ISH. The absolute number of ICP4 and TK transcripts in KOS-infected ganglia increased dramatically (by about 1000-fold) during the next 40 h, while viral DNA synthesis took place, whereas it rose only 10-fold during the same period in TK^- mutant-infected ganglia. Throughout the time period studied, the number of ICP4 and TK transcripts *per viral genome* remained essentially constant for ganglia infected with mutant and wild-type virus. Quantitatively, the results show that the levels of ICP4 and TK RNAs in neurons are proportional to viral genome copy-number throughout infection, as opposed to the situation in tissue-culture cells where genome amplification does not result in an increase in the accumulation of these transcripts. The hypothesis of altered regulation in neurons suggests that the latent genome is responsive to signals that result in DNA replication, through increases in IE protein levels or, perhaps, via direct activation of early transcription.

Following from the above studies with TK^- mutants, if viral DNA replication is an important control point it might be predicted that IE and early transcripts would be present, at low levels, during latency. Analysis by sensitive RT–PCR has shown this to be the case for ICP4 and TK RNAs in one study (Kramer & Coen, 1995), and for ICP27 and ICP6 RNAs in another (Tal-Singer *et al*., 1997). If these findings signify a general low-level production of nonlatent (by the classical definition) transcripts in many neurons, a post-IE block may indeed be the crucial barrier to virus replication.

Studies on gene expression in infected cultured foetal rat DRG neurons, analogous to the cultures used in the latency model of Wilcox *et al*. discussed above, support the concept that DNA synthesis is a trigger for IE and early gene expression in neurons (Nichol *et al*., 1996). Compared with permissive Vero cells, infected DRG neurons accumulated low levels of IE and early transcripts if viral DNA synthesis was inhibited, just as described for ganglia from mice (Kosz-Vnenchak *et al*., 1990). Without inhibitors the levels of IE and early transcripts were initially low, but they increased dramatically once viral DNA synthesis commenced. The triggering effect was attributed to both the initiation and elongation stages of DNA synthesis, and it was noted that IE genes, especially that encoding ICP4, are located close to the short region origin of replication (Nichol *et al*., 1996). Thus, the early steps in DNA synthesis may induce structural changes that activate transcription of the ICP4 gene. Nichol *et al*. (1996) further speculated that synthesis of the *UL9* product, the main origin-binding protein, may be induced by stimuli that promote reactivation, since the *UL9* promoter responds to cAMP in PC12 cells (Deb *et al*., 1994). The studies with cultured neurons therefore suggest that it is possible for reactivation to occur through pathways which lead directly to DNA replication.

Further evidence for unusual gene regulation in neurons was obtained when the time-course of transcript expression after explantation of ganglia was monitored by RT–PCR (Tal-Singer *et al*., 1997). Viral RNAs specific for TK and ICP6, as well as for the late protein VP5, were detected earlier than IE-specific transcripts (including that encoding ICP0). The first transcripts to be detected in increased amounts were those specified by the proto-oncogenes c-*fos* and c-*jun*, leading the authors to suggest that cellular transactivators of this type (or the factors that induce them) may turn on viral early gene expression and bypass the requirement for IE proteins. The net effect of increasing early gene expression could be activation of DNA replication and consequent stimulation of IE transcription. The thrust of this idea, that cellular factors stimulate expression from the latent genome, is similar to that of studies discussed above (Cai & Schaffer, 1991; Ralph *et al*., 1994; Jordan *et al*., 1998). There is, however, a difference in emphasis: the hypothesis of P. A. Schaffer's group supposes that the cellular proteins function in a manner analogous to that of ICP0, whereas the report of Tal-Singer *et al*. (1997) views the latent genome as intrinsically sensitive to cellular signalling pathways that are activated in response to reactivation stimuli, possibly circumventing a prior requirement for IE gene expression.

Investigation of the intracellular localization of HCF demonstrated that stimuli which induce reactivation of latent HSV also result in the transport of HCF from the cytoplasm to the nucleus of sensory neurons (Kristie *et al*., 1999). It was proposed that this event may activate IE transcription, again presupposing that the latent genome is inherently available for the action of transactivator proteins.

The results described in this section are taken from a range of experiments, but putting them together it is possible to view latency as arising from an initially inefficient production of IE proteins in neurons which results in subsequent gene expression falling below a critical level needed for the initiation of DNA synthesis. From this standpoint, it is not necessary to postulate conversion of the genome to a repressed state, and more straightforward to consider that promoters are potentially responsive to cellular and viral transcription and replication factors. More compellingly, reactivation need not require the synthesis of ICP0 or a cellular functional homologue. Changes in the profile of transcription factors, through synthesis of new proteins or activation of signal transduction pathways, could suffice to turn on the expression of IE or early viral genes.

7. Comparisons

The preceding discussion outlines two apparently conflicting views on the functional state of the latent genome. The concepts in the 'repressed genome' model are derived from studies on tissue-culture cells. The 'responsive genome' model originates from various approaches but there is a chain of cohesion: the mouse experiments of Kosz-Vnenchak *et al*. (1990) are supported by those of Nichol *et al*. (1996) in cultured neurons, which in turn deal with the early events in the latency system of C. L. Wilcox and colleagues. Although direct comparative experiments to distinguish the models have not been performed, there are two observations that bring the differences into focus. First, latency in cultured DRG neurons is, in general, less stable than quiescence in fibroblasts, with reactivation occurring readily in response to stimuli such as activation of protein kinase C and cyclic AMP-mediated pathways (Smith *et al*., 1992). Agents that mediate these effects do not disrupt the quiescent state in fibroblasts (C. M. Preston, unpublished observations). In one specific example from published studies, treatment of latently infected neuronal cultures with cycloheximide for 1 h resulted in reactivation of virus from most neurons even in the presence of NGF (Wilcox *et al*., 1990), whereas addition of this inhibitor to fibroblasts for 6 h had no effect on quiescent genomes of IE-deficient mutants (Preston & Nicholl, 1997). Second, there is a major difference between DRG neuron and fibroblast latency models in the requirement for ICP0: the absence of this protein results in lower efficiency of establishment in neuronal cultures, whereas the presence of ICP0 prevents the attainment of the quiescent state in fibroblasts. Therefore, in the model using foetal DRG neurons, expression of ICP0 enhances the establishment of latency, and reactivation is achieved by transient cellular changes which appear less dramatic than required in fibroblasts.

8. A late block?

Although most observations support the concept that an early decision determines the outcome of infection in neurons, there are data that apparently do not fit with this view. The high copy number of viral genomes during latency has long been a concern for models in which no viral gene expression is envisaged, especially as the nonlinear form of the genome could represent replicated concatemeric DNA that is not packaged into virions, as well as circularized input DNA. In some systems, infection with mutants that do not replicate in neurons (such as TK^- mutants) results in the retention of fewer genomes during latency (Efstathiou *et al*., 1989; Sawtell, 1997; Bates & DeLuca, 1998; Kramer *et al*., 1998). This may well be due entirely to differential loading of neurons from the periphery, but it is difficult to dismiss the suspicion that some latent genomes are retained after viral DNA replication has occurred *in the neuron*. In some systems where TK^- viruses have been compared with wild-type counterparts, the reduced accumulation of DNA is matched by a correspondingly lower level of LATs (and number of LAT^+ neurons) such that the ratio of LATs to DNA is unchanged (Bates & DeLuca, 1998; Kramer *et al*., 1998). The simplest conclusion from this result is that, in the absence of replication in the neuron, there are fewer latently infected cells due to reduced virus spread but with equivalent genome copy numbers, which is compatible with a pre-replication block to gene expression. In a mouse flank infection model, however, the neurons of the ganglia directly innervating the site of inoculation, which support acute phase replication, retained a 10-fold higher latent viral genome copy number per LAT^+ cell than the neurons in adjacent ganglia, in which replication did not detectably occur (Speck & Simmons, 1991; Simmons *et al*., 1992; Slobedman *et al*., 1994). After infection with a TK^- mutant, all ganglia examined contained the lower number of genomes per LAT^+ neuron characteristic of adjacent ganglia in mice infected with wild-type HSV, suggesting that the neurons with high copy number retained some replicated progeny DNA (Slobedman *et al*., 1994). It should be cautioned, however, that the TK^- mutant used is impaired for replication at the site of inoculation, probably due to inefficient production of the UL24 gene product, and that the reduced amount of viral DNA in ganglia may be due to this defect rather than the inability to replicate the genome (Coen *et al*., 1989; Efstathiou *et al*., 1989). Qualitatively, explant reactivation of wild-type HSV-1 across the spectrum of ganglia tested in the flank system correlated with expression of LATs rather than viral DNA per LAT^+ neuron, suggesting that the extra genomes represent a 'dead end' (Speck & Simmons, 1991; Simmons *et al*., 1992). A more quantitative analysis showed that, after ear inoculation of mice, reactivation (measured as gene expression from the ICP0 promoter) is more efficient in ganglia innervating the site of inoculation than in adjacent ganglia (Lachmann *et al*., 1999). If the observations on genome copy-number distribution in the flank inoculation system are valid for the ear model, these studies suggest that progeny genomes may persist *and be competent for reactivation*. Support for this view comes from the finding that copy-number per infected neuron, and not the proportion of infected neurons, correlates with the different *in vivo* reactivation frequencies of HSV-1 strains 17 and KOS, although the reason for the different retention of DNA by the two strains is not known (Sawtell *et al*., 1998).

9. Questions

It is clear from the above text that there are still many uncertainties regarding the mechanism of gene regulation during latency. The following questions focus on some of the issues that remain to be resolved.

(i) What is the relevance of the quiescent state reached in tissue-culture cells to latency *in vivo*? Is it reasonable to extrapolate from the events that occur in fibroblasts, on the basis that the observed repression is an inbuilt cellular mechanism for gene silencing, or is the counter argument, that the HSV genome is controlled by the specialized transcriptional status of the neuron, correct? Despite the fact that most infected neurons are overtly LAT$^-$ *in vivo*, expression of LATs is considered to be a hallmark of latency and it has not yet been possible to set up a tissue-culture system in which IE-impaired mutants become quiescent yet continue to produce LATs. Studies in fibroblasts have given information on gene repression and the mode of action of ICP0, but there is still little convincing evidence that the findings are directly relevant to latency *in vivo*. Until tissue-culture studies make useful predictions that are tested and verified in animal-model systems they will remain vulnerable to the criticism, made in two widely quoted reviews, that they are irrelevant to latency and useful only for the understanding of gene expression in nonpermissive cells (Roizman & Sears, 1987; Stevens, 1989).

(ii) Do studies on cultured neurons reproduce interactions that occur *in vivo*? The elegance of using the natural host cell for studies of HSV latency does not protect the system of C. L. Wilcox and colleagues from the suggestion that it is artificial. For optimal performance it is necessary to inhibit viral DNA synthesis, yet this block per se may result in an unusual interaction in cells that would otherwise be killed. Reactivation may be easy to accomplish by alteration of signalling pathways just because the imposed block is relatively late in the programme of viral gene expression. NGF-dependent latency *can* be established without acycloguanosine, but the efficiency of reactivation is low: cultures of 5000 neurons were each infected with 150 p.f.u. of HSV-1, but NGF withdrawal induced reactivation from only 10% of surviving cultures (Wilcox & Johnson, 1987). Does this mean that most of the initially applied genomes under these conditions are not retained during latency, that they are not competent for reactivation by NGF withdrawal, or is it a reasonable finding in view of the low efficiency of reactivation *in vivo*? The data of Wilcox *et al*. (1997) clearly show that the establishment of latency is greatly reduced when ICP0 is absent, in contrast to the situation in fibroblasts. This difference is a benchmark for the relevance of the different culture systems; thus it is imperative that experiments to determine whether ICP0 is required for efficient establishment of latency *in vivo*, as distinct from the role of the protein in replication at the site of inoculation, be devised.

(iii) Does a block at the level of viral DNA synthesis *in vivo* represent a relevant way of arresting the virus replication cycle? The experiments of Kosz-Vnenchak *et al*. (1990), Nichol *et al*. (1996) and Kramer *et al*. (1998) show that the levels of IE and early transcripts rise when viral genome replication occurs, but it does not necessarily follow that a natural control is exerted at this point. A vital issue concerns the nature of the population of genomes which give rise to IE and early transcripts in neurons of mice infected with TK$^-$ mutants. Since the transcripts are almost undetectable by ISH it is not possible to determine whether they are present in as many cells as progress to replication in wild-type virus infections, or if they represent a subpopulation that is destined for latency. The data of Kramer *et al*. (1998) showed no difference in the levels of IE and early transcripts between TK$^-$ and wild-type virus infections at times prior to the commencement of DNA replication. If these transcripts arise *totally* from genomes destined for replication and elimination (but not achieving this fate in the absence of DNA synthesis), there is no need to postulate that the altered gene expression pattern in neurons is related to latency. The levels of transcripts at early times after infection with wild-type and TK$^-$ mutants are greater than those retained during latency with wild-type HSV-1 (latency after infection with the TK$^-$ mutant was, unfortunately, not tested), so either many neurons expressing them die or the initial gene expression declines (Kramer *et al*., 1998).

(iv) What is the origin of the IE and early transcripts detected in murine ganglia during latency? Are these indicative of low-level transcription in most latently infected neurons, in apparent contradiction of models envisaging global repression of the viral genome? Do they represent a subpopulation blocked at a stage later than IE transcription but prior to DNA synthesis, as predicted by the hypothesis of M. Kosz-Vnenchak and colleagues? Are they simply remnants of an initial lytic infection or an abortive reactivation in one or two neurons, in which cases the transcripts do not bear on the issue of gene control during latency? The application of sophisticated techniques, such as *in situ* RT–PCR or RT–PCR combined with CXA, will be necessary to resolve this issue.

(v) What is the molecular basis of the intracellular changes that promote reactivation? This is crucial for distinguishing between the various models for gene repression during latency, yet it is the least well understood aspect of the entire subject. If reactivation occurs through cellular factors that act in the same way as ICP0, it is reasonable to consider that the repression observed in fibroblasts, when IE transcription or

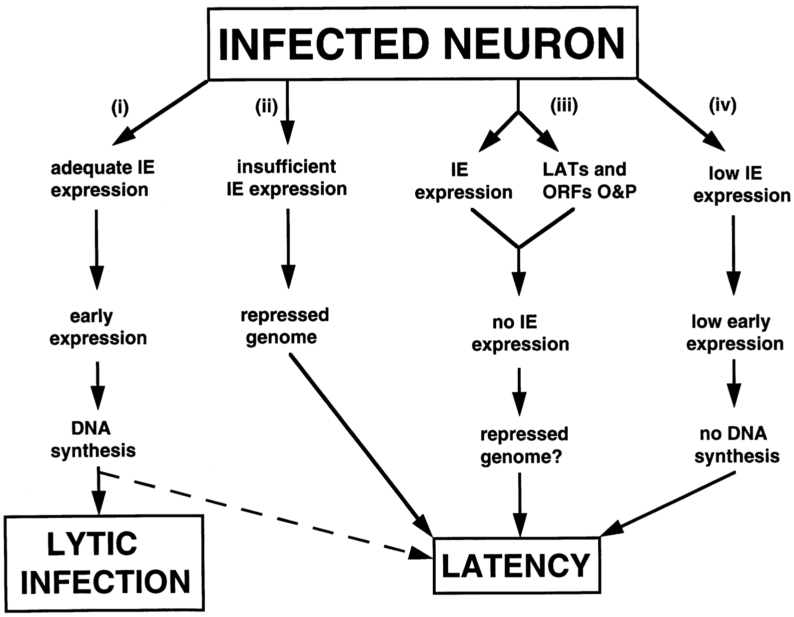

Fig. 2. Diagrammatic representation of possible outcomes of infection of neurons. The four possible pathways [(i) to (iv)] proposed in the text are outlined.

ICP0 function are inhibited, also occurs in neurons and accounts for the silence of the genome during latency. If, however, reactivation is due to changes which require only the activation of IE or early promoters rather than a major reversal of genome repression, the results from cultured neurons should be considered the more relevant. As a further variation, it may be that the ICP0 promoter is uniquely sensitive to activation by cellular transcription factors in latently infected neurons, such that production of ICP0 precedes unblocking of the remainder of the genome. It should be noted, though, that the ICP0 promoter is repressed in fibroblasts and is not detectably active *in vivo* (Preston & Nicholl, 1997; Tal-Singer *et al*., 1997; Lachmann *et al*., 1999). A further complexity is that, in view of the inefficiency of reactivation, only a subpopulation of neurons, or of genomes within a given neuron, may be responsive to inducing signals. If, as suggested by the data of Sawtell *et al*. (1998), high viral DNA copy-number predisposes ganglia to reactivation, it may be that resumption of transcription by a single genome that is not fully repressed initiates a chain reaction: once ICP0 has been synthesized, other genomes that were fully repressed could be switched on.

(vi) How is the latent genome organized? Is it in a tightly packed structure, such as heterochromatin, that prevents access of transactivator proteins? If so, does the structure apply to the entire genome or to specific regions? Study of this aspect of the latent genome *in vivo* will be very demanding, and it may be preferable first to resolve this question in a cell culture system.

10. Conclusion

It will be clear that there is much to be done before the molecular basis for latency is understood. I offer the following interpretation of the available data, shown diagrammatically in Fig. 2, to stimulate further research. For whatever reason, IE transcription is inefficient in neurons and four interactions may ensue.

(i) Some viral genomes produce sufficient amounts of IE proteins to support productive replication in the neuron, and these are subsequently cleared by the host. There is no doubt that this occurs.

(ii) Many genomes do not produce ICP0 quickly enough and are repressed in a manner analogous to that seen in fibroblasts. It seems to me unlikely that such a general outcome of infection in the absence of IE gene expression does not occur in neurons, especially when considering the similarities between quiescence and latency and the great difficulties experienced in obtaining long-term neuronal gene expression from HSV vectors.

(iii) Some genomes escape the initial repression, but IE gene expression and function (especially that of ICP0) is inhibited by the action of LATs and/or the ORF O and P products, so these genomes also become silenced. This classifies LATs and ORFs O and P as 'safety valves', which suppress IE gene expression when the initial block to IE transcription has been evaded.

(iv) A population of genomes escapes the control points of (ii) and (iii), and is blocked at a stage later than IE gene expression but prior to (or possibly after) DNA synthesis, resulting in an interaction from which reactivation can be induced by changes to neuronal physiology, as in cultured neurons. Whether this class of latent genome exists *in vivo* is, to me, the most difficult question to resolve. How the variations in viral DNA content, expression of LATs and reactivation competence that are observed in individual neurons are superimposed on the possible pathways to latency are questions for the future.

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References

Ace, C. I., McKee, T. A., Ryan, J. M., Cameron, J. M. & Preston, C. M. (1989). Construction and characterization of a herpes simplex virus type 1 mutant unable to transinduce immediate-early gene expression. *Journal of Virology* **63**, 2260–2269.

Alvira, M. R., Goins, W. F., Cohen, J. B. & Glorioso, J. G. (1999). Genetic studies exposing the splicing events involved in herpes simplex virus type 1 latency-associated transcript production during lytic and latent infection. *Journal of Virology* **73**, 3866–3876.

Arthur, J. L., Everett, R. D., Brierley, I. & Efstathiou, S. (1998). Disruption of the 5′ and 3′ splice sites flanking the major latency-associated transcripts of herpes simplex type 1: evidence for alternate splicing in lytic and latent infections. *Journal of General Virology* **79**, 107–116.

Bates, P. A. & DeLuca, N. A. (1998). The polyserine tract of herpes simplex virus ICP4 is required for normal viral gene expression and growth in murine trigeminal ganglia. *Journal of Virology* **72**, 7115–7124.

Bohensky, R. A., Papavassiliou, A. G., Gelman, I. H. & Silverstein, S. (1993). Identification of a promoter mapping within the reiterated sequences that flank the herpes simplex virus type 1 UL region. *Journal of Virology* **67**, 632–642.

Bohensky, R. A., Lagunoff, M., Roizman, B., Wagner, E. K. & Silverstein, S. (1995). Two overlapping transcription units which extend across the L–S junction of herpes simplex virus type 1. *Journal of Virology* **69**, 2889–2897.

Bruni, R. & Roizman, B. (1996). Open reading frame P – a herpes simplex virus gene repressed during productive infection encodes a protein that binds a splicing factor and reduces synthesis of viral proteins made from spliced mRNA. *Proceedings of the National Academy of Sciences, USA* **93**, 10423–10427.

Cai, W. & Schaffer, P. A. (1989). Herpes simplex virus type 1 ICP0 plays a critical role in the de novo synthesis of infectious virus following transfection of viral DNA. *Journal of Virology* **63**, 4579–4589.

Cai, W. & Schaffer, P. A. (1991). A cellular function can enhance gene expression and plating efficiency of a mutant defective in the gene for ICP0, a transactivating protein of herpes simplex virus type 1. *Journal of Virology* **65**, 4078–4090.

Cai, W. & Schaffer, P. A. (1992). Herpes simplex virus type 1 ICP0 regulates expression of immediate-early, early and late genes in productively infected cells. *Journal of Virology* **66**, 2904–2915.

Cai, W., Astor, T. L., Liptak, L. M., Cho, C., Coen, D. M. & Schaffer, P. A. (1993). The herpes simplex virus type 1 regulatory protein ICP0 enhances viral replication during acute infection and reactivation from latency. *Journal of Virology* **67**, 7501–7512.

Carrozza, M. J. & DeLuca, N. A. (1996). Interaction of the viral activator protein ICP4 with TFIID through TAF250. *Molecular and Cellular Biology* **16**, 3085–3093.

Chelbi-Alix, M. K. & de The, H. (1999). Herpes virus induced proteasome-dependent degradation of the nuclear bodies-associated PML and Sp100 proteins. *Oncogene* **18**, 935–941.

Chen, J. & Silverstein, S. (1992). Herpes simplex viruses with mutations in the gene encoding ICP0 are defective in gene expression. *Journal of Virology* **66**, 2916–2927.

Chen, X., Schmidt, M. C., Goins, W. F. & Glorioso, J. C. (1995). Two herpes simplex virus type 1 latency-active promoters differ in their contributions to latency-associated transcript expression during lytic and latent infections. *Journal of Virology* **69**, 7899–7908.

Chen, S.-H., Kramer, M. F., Schaffer, P. A. & Coen, D. M. (1997). A viral function represses accumulation of transcripts from productive-cycle genes in mouse ganglia latently infected with herpes simplex virus. *Journal of Virology* **71**, 5878–5884.

Cheung, P., Panning, B. & Smiley, J. R. (1997). Herpes simplex virus immediate-early proteins ICP0 and ICP4 activate the endogenous human α-globin gene in nonerythroid cells. *Journal of Virology* **71**, 1784–1793.

Cleary, M. A., Stern, S., Tanaka, M. & Herr, W. (1993). Differential positive control by Oct-1 and Oct-2 – activation of a transcriptionally silent motif through Oct-1 and VP16 corecruitment. *Genes & Development* **7**, 72–83.

Clements, G. B. & Stow, N. D. (1989). A herpes simplex virus type 1 mutant containing a deletion within immediate early gene 1 is latency-competent in mice. *Journal of General Virology* **70**, 2501–2506.

Coen, D. M., Weinheimer, S. P. & McKnight, S. L. (1986). A genetic approach to promoter recognition during trans induction of viral gene expression. *Science* **234**, 53–59.

Coen, D. M., Kosz-Vnenchak, M., Jacobson, J. G., Leib, D. A., Bogard, C. L., Schaffer, P. A., Tyler, K. L. & Knipe, D. M. (1989). Thymidine kinase-negative herpes simplex virus mutants establish latency in mouse trigeminal ganglia but do not reactivate. *Proceedings of the National Academy of Sciences, USA* **86**, 4736–4740.

Cook, W. J., Lin, S. M., DeLuca, N. A., Moyinhan, E. B. & Coen, D. M. (1995). Induction of transcription by a viral regulatory protein depends on the relative strengths of functional TATA boxes. *Molecular and Cellular Biology* **15**, 4998–5006.

Deb, S. P., Deb, S. & Brown, D. R. (1994). Cell-type-specific induction of the UL9 gene of HSV-1 by cell signalling pathway. *Biochemical and Biophysical Research Communications* **205**, 44–51.

DeLuca, N. A. & Schaffer, P. A. (1985). Activation of immediate-early, early and late promoters by temperature-sensitive and wild-type forms of herpes simplex virus type 1 protein ICP4. *Molecular and Cellular Biology* **5**, 558–570.

Dent, C. L., Lillycrop, K. A., Estridge, J. K., Thomas, N. S. B. & Latchman, D. S. (1991). The B-cell and neuronal forms of the octamer-binding protein Oct-2 differ in DNA-binding specificity and functional activity. *Molecular and Cellular Biology* **11**, 3925–3930.

Deshmane, S. L. & Fraser, N. W. (1989). During latency, herpes simplex virus type 1 DNA is associated with nucleosomes in a chromatin structure. *Journal of Virology* **63**, 943–947.

Devi-Rao, G. B., Goodart, S. A., Hecht, L. M., Rochford, R., Rice, M. A. & Wagner, E. K. (1991). Relationship between polyadenylated and nonpolyadenylated herpes simplex virus type 1 latency-associated transcripts. *Journal of Virology* **65**, 2179–2190.

Devi-Rao, G. B., Bloom, D. C., Stevens, J. G. & Wagner, E. K. (1994). Herpes simplex virus type 1 DNA replication and gene expression during explant-induced reactivation of latently infected murine sensory ganglia. *Journal of Virology* **68**, 1271–1282.

Dobson, A. T., Margolis, T. P., Sedarati, F., Stevens, J. G. & Feldman, L. T. (1990). A latent, nonpathogenic HSV-1-derived vector stably expresses β-galactosidase in mouse neurons. *Neuron* **5**, 353–360.

Dolan, A., Jamieson, F. E., Cunningham, C., Barnett, B. C. & McGeoch, D. J. (1998). The genome sequence of herpes simplex virus type 2. *Journal of Virology* **72**, 2010–2021.

Ecob-Prince, M. S. & Hassan, K. (1994). Reactivation of latent herpes simplex virus from explanted dorsal root ganglia. *Journal of General Virology* **75**, 2017–2028.

Ecob-Prince, M. S., Preston, C. M., Rixon, F. J., Hassan, K. & Kennedy, P. G. E. (1993*a*). Neurons containing latency-associated transcripts are numerous and widespread in dorsal root ganglia following footpad inoculation of mice with herpes simplex virus type 1 mutant *in*1814. *Journal of General Virology* **74**, 985–994.

Ecob-Prince, M. S., Rixon, F. J., Preston, C. M., Hassan, K. & Kennedy, P. G. E. (1993*b*). Reactivation *in vivo* and *in vitro* of herpes simplex virus from mouse dorsal root ganglia which contain different levels of latency-associated transcripts. *Journal of General Virology* **74**, 995–1002.

Efstathiou, S., Minson, A. C., Field, H. J., Anderson, J. R. & Wildy, P. (1986). Detection of herpes simplex virus-specific DNA sequences in latently infected mice and humans. *Journal of Virology* **57**, 446–455.

Efstathiou, S., Kemp, S., Darby, G. K. & Minson, A. C. (1989). The role of herpes simplex virus type 1 thymidine kinase in pathogenesis. *Journal of General Virology* **70**, 869–879.

Eissenberg, J. C., Morris, G., Reuter, G. & Harnett, T. (1992). The heterochromatin associated protein HP1 is an essential protein in *Drosophila* with dosage effects on position effect variegation. *Genetics* **131**, 345–352.

Everett, R. D. (1984). Transactivation of transcription by herpes virus products: requirement for two HSV-1 immediate-early polypeptides for maximum activity. *EMBO Journal* **3**, 3135–3141.

Everett, R. D. (1985). Activation of cellular promoters during herpes simplex virus infection of biochemically transformed cells. *EMBO Journal* **4**, 1973–1980.

Everett, R. D. (1989). Construction and characterization of herpes simplex virus type 1 mutants with defined lesions in immediate early gene 1. *Journal of General Virology* **70**, 1185–1202.

Everett, R. D. & Maul, G. G. (1994). HSV-1 IE protein Vmw110 causes redistribution of PML. *EMBO Journal* **13**, 5062–5069.

Everett, R. D., Freemont, P., Saitoh, H., Orr, A., Kathoria, M. & Parkinson, J. (1998*a*). The disruption of ND10 during herpes simplex virus infection correlates with the Vmw110 and proteasome-dependent loss of several PML isoforms. *Journal of Virology* **72**, 6581–6591.

Everett, R. D., Orr, A. & Preston, C. M. (1998*b*). A viral activator of gene expression functions via the ubiquitin-proteasome pathway. *EMBO Journal* **17**, 7161–7169.

Everett, R. D., Earnshaw, W. C., Findlay, J. & Lomonte, P. (1999*a*). Specific destruction of kinetochore protein CENP-C and disruption of cell division by herpes simplex virus immediate-early protein Vmw110. *EMBO Journal* **18**, 1526–1538.

Everett, R. D., Earnshaw, W. C., Pluta, A. F., Sternsdorf, T., Ainsztein, A. M., Carmena, M., Ruchaud, S., Hsu, W.-L. & Orr, A. (1999*b*). A dynamic connection between centromeres and ND10 proteins. *Journal of Cell Science* **112**, 3443–3454.

Farrell, M. J., Dobson, A. T. & Feldman, L. T. (1991). Herpes simplex virus latency-associated transcript is a stable intron. *Proceedings of the National Academy of Sciences, USA* **88**, 790–794.

Feldman, L. T. (1994). Transcription of the HSV-1 genome in neurons *in vivo*. *Seminars in Virology* **5**, 207–212.

Field, H. J. & Wildy, P. W. (1978). The pathogenicity of thymidine kinase-deficient mutants of herpes simplex virus in mice. *Journal of Hygiene* **81**, 267–277.

Fraser, N. W., Block, T. M. & Spivack, J. G. (1992). The latency-associated transcripts of herpes simplex virus: RNA in search of a function. *Virology* **191**, 1–8.

Garber, D. A., Schaffer, P. A. & Knipe, D. M. (1997). A LAT-associated function reduces productive-cycle gene expression during acute infection of murine sensory neurons with herpes simplex virus type 1. *Journal of Virology* **71**, 5885–5893.

Glorioso, J. C., Goins, W. F. & Fink, D. J. (1992). Herpes simplex virus-based vectors. *Seminars in Virology* **3**, 265–276.

Glorioso, J. C., DeLuca, N. A. & Fink, D. J. (1995). Development and application of herpes simplex virus vectors for human gene therapy. *Annual Review of Microbiology* **49**, 675–710.

Gordon, J. Y., McKnight, J. L., Ostrove, J. M., Romanowski, E. & Araullo-Cruz, T. (1990). Host species and strain differences affect the ability of an HSV-1 ICP0 deletion mutant to establish latency and spontaneously reactivate *in vivo*. *Virology* **178**, 469–477.

Gressens, P. & Martin, J. R. (1994). In situ polymerase chain reaction: localization of HSV-2 DNA sequences in infections of the nervous system. *Journal of Virological Methods* **46**, 61–83.

Hagmann, M., Georgiev, O., Schaffner, W. & Douville, P. (1995). Transcription factors interacting with herpes simplex virus α gene promoters in sensory neurons. *Nucleic Acids Research* **23**, 4978–4985.

Halford, W. P., Gebhardt, B. M. & Carr, D. J. J. (1996). Mechanisms of herpes simplex virus type 1 reactivation. *Journal of Virology* **70**, 5051–5060.

Harris, R. A. & Preston, C. M. (1991). Establishment of latency *in vitro* by the herpes simplex virus type 1 mutant *in*1814. *Journal of General Virology* **72**, 907–913.

He, X., Treacy, M. N., Simmons, D. M., Ingraham, H. A., Swanson, L. S. & Rosenberg, M. G. (1989). Expression of a large family of POU-domain relatory genes in mammalian brain development. *Nature* **340**, 35–42.

Herr, W. & Cleary, M. A. (1995). The POU domain: versatility in transcriptional regulation by a flexible two-in-one DNA-binding domain. *Genes & Development* **9**, 1679–1693.

Honess, R. W. & Roizman, B. (1974). Regulation of herpesvirus macromolecular synthesis. I. Cascade regulation of the synthesis of three groups of viral proteins. *Journal of Virology* **14**, 8–19.

James, T. C., Eissenberg, J. C., Craig, C., Deitrich, V., Hobson, A. & Elgin, S. C. R. (1989). Distribution patterns of HP1, a heterochromatin-associated nonhistone chromosomal protein of *Drosophila*. *European Journal of Cell Biology* **50**, 170–180.

Jamieson, D. R. S., Robinson, L. H., Daksis, J. I., Nicholl, M. J. & Preston, C. M. (1995). Quiescent viral genomes in human fibroblasts after infection with herpes simplex virus Vmw65 mutants. *Journal of General Virology* **76**, 1417–1431.

Johnson, P. A., Miyanohara, A., Levine, F., Cahill, T. & Friedmann, T. (1992). Cytotoxicity of a replication defective mutant of herpes simplex virus type 1. *Journal of Virology* **66**, 2952–2965.

Johnson, P. A., Wang, M. J. & Friedmann, T. (1994). Improved cell survival by the reduction of immediate-early gene expression in replication-defective mutants of herpes simplex virus type 1 but not by mutation of the virion host shutoff function. *Journal of Virology* **68**, 6347–6362.

Jordan, R., Pepe, J. & Schaffer, P. A. (1998). Characterization of a nerve growth factor-inducible cellular activity that enhances herpes simplex virus type 1 gene expression and replication of an ICP0 null mutant in cells of neural lineage. *Journal of Virology* **72**, 5373–5382.

Katz, J. P., Bodin, E. T. & Coen, D. M. (1990). Quantitative polymerase chain reaction analysis of herpes simplex virus DNA in ganglia of mice infected with replication-incompetent mutants. *Journal of Virology* **64**, 4288–4295.

Kemp, L. M. & Latchman, D. S. (1989). Regulated expression of herpes simplex virus immediate-early genes in neuroblastoma cells. *Virology* **171**, 607–610.

Kemp, L. M., Dent, C. L. & Latchman, D. S. (1990). Octamer motif mediates transcriptional repression of HSV immediate-early genes and octamer-containing cellular promoters in neuronal cells. *Neuron* **4**, 215–222.

Kosz-Vnenchak, M., Coen, D. M. & Knipe, D. M. (1990). Restricted expression of herpes simplex virus lytic genes during establishment of latent infection by thymidine kinase-negative mutant viruses. *Journal of Virology* **64**, 5396–5402.

Kosz-Vnenchak, M., Jacobsen, J., Coen, D. M. & Knipe, D. M. (1993). Evidence for a novel regulatory pathway for herpes simplex virus gene expression in trigeminal ganglion neurons. *Journal of Virology* **67**, 5383–5393.

Kramer, M. F. & Coen, D. M. (1995). Quantification of transcripts from the ICP4 and thymidine kinase genes in mouse trigeminal ganglia latently infected with herpes simplex virus. *Journal of Virology* **69**, 1389–1399.

Kramer, M. F., Chen, S.-H., Knipe, D. M. & Coen, D. M. (1998). Accumulation of viral transcripts and DNA during establishment of latency by herpes simplex virus. *Journal of Virology* **72**, 1177–1185.

Kristie, T. M. & Roizman, B. (1988). Differentiation and DNA contact points of the host proteins binding at the *cis* site for virion-mediated induction of herpes simplex virus 1 α genes. *Journal of Virology* **62**, 1145–1157.

Kristie, T. M., Pomerantz, J. L., Twomey, T. C., Parent, S. A. & Sharp, P. A. (1995). The cellular C1 factor of the herpes simplex virus enhancer complex is a family of polypeptides. *Journal of Biological Chemistry* **270**, 4387–4394.

Kristie, T. M., Vogel, J. L. & Sears, A. E. (1999). Nuclear localization of the C1 factor (host cell factor) in sensory neurons correlates with reactivation of herpes simplex virus from latency. *Proceedings of the National Academy of Sciences, USA* **96**, 1229–1233.

Kwon, B. S., Gangarosa, L. P., Burch, K. D., Deback, J. & Hill, J. M. (1981). Induction of ocular herpes simplex virus shedding induced by iontophoresis of epinephrine into rabbit cornea. *Investigative Ophthalmology & Visual Science* **21**, 442–449.

La Boissière, S., Hughes, T. & O'Hare, P. (1999). HCF-dependent nuclear import of VP16. *EMBO Journal* **18**, 480–489.

Lachmann, R. H. & Efstathiou, S. (1997). Utilization of the herpes simplex virus type 1 latency-associated regulatory region to drive stable reporter gene expression in the nervous system. *Journal of Virology* **71**, 3197–3207.

Lachmann, R. H., Browne, H. C. & Efstathiou, S. (1996). A murine RNA polymerase I promoter inserted into the herpes simplex virus type 1 genome is functional during lytic, but not latent, infection. *Journal of General Virology* **77**, 2575–2582.

Lachmann, R. H., Sadarangani, M., Atkinson, H. R. & Efstathiou, S. (1999). An analysis of herpes simplex virus gene expression during latency establishment and reactivation. *Journal of General Virology* **80**, 1271–1282.

Lagunoff, M. & Roizman, B. (1994). Expression of a herpes simplex virus 1 open reading frame antisense to the $_{\gamma1}$34.5 gene and transcribed by an RNA 3′ coterminal with the unspliced latency-associated transcript. *Journal of Virology* **68**, 6021–6028.

Lagunoff, M. & Roizman, B. (1995). The regulation of synthesis and properties of the protein product of open reading frame P of the herpes simplex virus 1 genome. *Journal of Virology* **69**, 3615–3623.

Lagunoff, M., Randall, G. & Roizman, B. (1996). Phenotypic properties of herpes simplex virus 1 containing a derepressed open reading frame P gene. *Journal of Virology* **70**, 1810–1817.

Latchman, D. S. (1999). POU family transcription factors in the nervous system. *Journal of Cellular Physiology* **179**, 126–133.

Lee, L. Y. & Schaffer, P. A. (1998). A virus with a mutation in the ICP4-binding site in the L/ST promoter of herpes simplex virus type 1, but not a virus with a mutation in open reading frame P, exhibits cell-type-specific expression of $_{\gamma1}$34.5 transcripts and latency-associated transcripts. *Journal of Virology* **72**, 4250–4264.

Lehming, N., Le Saux, A., Schuller, J. & Ptashne, M. (1998). Chromatin components as part of a putative transcriptional repressing complex. *Proceedings of the National Academy of Sciences, USA* **95**, 7322–7326.

Leib, D. A., Coen, D. M., Bogard, C. L., Hicks, K. A., Yager, D. R., Knipe, D. M., Tyler, K. L. & Schaffer, P. A. (1989). Immediate-early regulatory gene mutants define different stages in the establishment and reactivation of herpes simplex virus latency. *Journal of Virology* **63**, 759–768.

Leist, T. P., Sandri-Goldin, R. M. & Stevens, J. G. (1989). Latent infections in spinal ganglia with thymidine kinase-deficient herpes simplex virus. *Journal of Virology* **63**, 4976–4978.

Lekstrom-Himes, J. A., Pesnicak, L. & Straus, S. E. (1998). The quantity of latent viral DNA correlates with the relative rates at which herpes simplex virus types 1 and 2 cause recurrent genital herpes outbreaks. *Journal of Virology* **72**, 2760–2764.

Lillycrop, K. A. & Latchman, D. S. (1992). Alternative splicing of the Oct-2 transcription factor RNA is differentially regulated in neuronal cells and B cells and results in protein isoforms with opposite effects on the activity of octamer/TAATGARAT-containing promoters. *Journal of Biological Chemistry* **267**, 24960–24965.

Lillycrop, K. A., Dent, C. L., Wheatley, S. C., Beech, N. M., Ninkina, N. N., Wood, J. N. & Latchman, D. S. (1991). The octamer-binding protein Oct-2 represses HSV immediate-early genes in cell lines derived from latently infectable sensory neurons. *Neuron* **7**, 381–390.

Lillycrop, K. A., Estridge, J. K. & Latchman, D. S. (1993). The octamer binding protein Oct-2 inhibits transactivation of the herpes simplex virus immediate-early genes by the virion protein Vmw65. *Virology* **196**, 888–891.

Lillycrop, K. A., Howard, M. K., Estridge, J. K. & Latchman, D. S. (1994). Inhibition of herpes simplex virus infection by ectopic expression of neuronal splice variants of the Oct-2 transcription factor. *Nucleic Acids Research* **22**, 815–820.

Lokensgard, J. R., Bloom, D. C., Dobson, A. T. & Feldman, L. T. (1994). Long-term promoter activity during herpes simplex virus latency. *Journal of Virology* **68**, 7148–7158.

Lokensgard, J. R., Berthomme, H. & Feldman, L. T. (1997). The latency-associated promoter of herpes simplex virus type 1 requires a region downstream of the transcription start site for long-term expression during latency. *Journal of Virology* **71**, 6714–6719.

McGeoch, D. J., Dalrymple, M. A., Davison, A. J., Dolan, A., Frame, M. C., McNab, D., Perry, L. J., Scott, J. E. & Taylor, P. (1988). The complete DNA sequence of the long unique region in the genome of herpes simplex virus type 1. *Journal of General Virology* **69**, 1531–1574.

McLennan, J. L. & Darby, G. (1980). Herpes simplex virus latency: the cellular location of virus in dorsal root ganglia and the fate of the infected cell following virus activation. *Journal of General Virology* **51**, 233–243.

Mador, N., Goldenberg, D., Cohen, O., Panet, A. & Steiner, I. (1998). Herpes simplex virus type 1 latency-associated transcripts suppress viral replication and reduce immediate-early gene mRNA levels in a neuronal cell line. *Journal of Virology* **72**, 5067–5075.

Maggioncalda, J., Mehta, A., Su, Y. H., Fraser, N. W. & Block, T. M. (1996). Correlation between herpes simplex virus type 1 reactivation from latent infection and the number of infected neurons in trigeminal ganglia. *Virology* **225**, 72–81.

Margolis, T. P., Sedarati, F., Dobson, A. T., Feldman, L. T. & Stevens, J. G. (1992). Pathways of viral gene expression during acute neuronal infection with HSV-1. *Virology* **189**, 150–160.

Maul, G. G. (1998). Nuclear domain 10, the site of DNA virus transcription and replication. *BioEssays* **20**, 660–667.

Maul, G. G. & Everett, R. D. (1994). The nuclear location of PML, a cellular member of the C_3HC_4 zinc-binding domain protein family, is rearranged during herpes simplex virus infection by the C_3HC_4 viral protein ICP0. *Journal of General Virology* **75**, 1223–1233.

Maul, G. G., Guldner, H. H. & Spivack, J. G. (1993). Modification of discrete nuclear domains induced by herpes simplex virus type 1 immediate early gene 1 product. *Journal of General Virology* **74**, 2679–2690.

Maul, G. G., Ishov, A. & Everett, R. D. (1996). Nuclear domain 10 as preexisting potential replication start sites of herpes simplex virus type-1. *Virology* **217**, 67–75.

Mehta, A., Maggioncalda, J., Bagasra, O., Thikkavarapu, S., Saikumari, P., Valyi-Nagy, T., Fraser, N. W. & Block, T. M. (1995). *In situ* PCR and RNA hybridization detection of herpes simplex virus sequences in trigeminal ganglia of latently infected mice. *Virology* **206**, 633–640.

Mellerick, D. M. & Fraser, N. W. (1987). Physical state of the latent herpes simplex virus genome in a mouse model system: evidence suggesting an episomal state. *Virology* **158**, 265–275.

Moriya, A., Yoshiki, A., Kita, M., Fushiki, S. & Imanishi, J. (1994). Heat shock-induced reactivation of herpes simplex virus type 1 in latently infected mouse trigeminal ganglion cells in dissociated culture. *Archives of Virology* **135**, 419–425.

Muller, S. & Dejean, A. (1999). Viral immediate-early proteins abrogate the modification by SUMO-1 of PML and Sp100 proteins, correlating with nuclear body disruption. *Journal of Virology* **73**, 5137–5143.

Nichol, P. F., Chang, J. Y., Johnson, E. M. & Olivo, P. D. (1996). Herpes simplex virus gene expression in neurons: viral DNA synthesis is a critical regulatory event in the branch point between the lytic and latent pathways. *Journal of Virology* **70**, 5476–5486.

O'Hare, P. (1993). The virion transactivator of herpes simplex virus. *Seminars in Virology* **4**, 145–155.

O'Hare, P. & Hayward, G. S. (1985). Evidence for a direct role for both the 175,000- and 110,000-molecular-weight immediate-early proteins of herpes simplex virus in the transactivation of delayed-early promoters. *Journal of Virology* **53**, 751–760.

Openshaw, H., Asher, L. V. S., Wohlenberg, C., Sekizawa, T. & Notkins, A. L. (1979). Acute and latent infection of sensory ganglia with herpes simplex virus: immune control and virus reactivation. *Journal of General Virology* **44**, 205–215.

Perng, G.-C., Ghiasi, H., Slanina, S., Nesburn, A. B. & Weschler, S. L. (1996). The spontaneous reactivation function of the herpes simplex virus type 1 LAT gene resides completely within the first 1·5 kilobases of the 8·3-kilobase primary transcript. *Journal of Virology* **70**, 976–984.

Phelan, A. & Clements, J. B. (1998). Posttranscriptional regulation in herpes simplex virus. *Seminars in Virology* **8**, 309–318.

Post, L. E. & Roizman, B. (1981). A generalized technique for deletion of specific genes in large genomes: α gene 22 of herpes simplex virus is not essential for growth. *Cell* **25**, 227–232.

Preston, C. M. (1979). Control of herpes simplex virus type 1 mRNA synthesis in cells infected with wild type virus or the temperature sensitive mutant tsK. *Journal of Virology* **29**, 275–284.

Preston, C. M. & Nicholl, M. J. (1997). Repression of gene expression upon infection of cells with herpes simplex virus type 1 mutants impaired for immediate early protein synthesis. *Journal of Virology* **71**, 7807–7813.

Preston, C. M., Mabbs, R. & Nicholl, M. J. (1997). Construction and characterization of herpes simplex virus type 1 mutants with conditional defects in immediate early gene expression. *Virology* **229**, 228–239.

Preston, C. M., Rinaldi, A. & Nicholl, M. J. (1998). Herpes simplex virus type 1 immediate early gene expression is stimulated by inhibition of protein synthesis. *Journal of General Virology* **79**, 117–124.

Puga, A., Rosenthal, J. D., Openshaw, H. & Notkins, A. L. (1978). Herpes simplex virus DNA and mRNA sequences in acutely and chronically infected trigeminal ganglia of mice. *Virology* **89**, 102–111.

Ralph, W. M., Cabatingan, M. S. & Schaffer, P. A. (1994). Induction of herpes simplex virus immediate-early gene expression by a cellular activity expressed in Vero and NB41A3 cells after growth arrest-release. *Journal of Virology* **68**, 6871–6882.

Ramakrishnan, R., Fink, D. J., Jiang, G., Desai, P., Glorioso, J. G. & Levine, M. (1994*a*). Competitive quantitative PCR analysis of herpes simplex virus type 1 DNA and latency-associated transcript RNA in latently infected cells of the rat brain. *Journal of Virology* **68**, 1864–1873.

Ramakrishnan, R., Levine, M. & Fink, D. J. (1994*b*). PCR-based analysis of herpes simplex virus type 1 latency in the rat trigeminal ganglion established with a ribonucleotide reductase-deficient mutant. *Journal of Virology* **68**, 7083–7091.

Ramakrishnan, R., Poliani, P. L., Levine, M., Glorioso, J. C. & Fink, D. J. (1996). Detection of herpes simplex virus type 1 latency-associated transcript expression in trigeminal ganglia by in situ reverse transcriptase PCR. *Journal of Virology* **70**, 6519–6523.

Randall, G., Lagunoff, M. & Roizman, B. (1997). The product of ORF O located within the domain of herpes simplex virus 1 genome transcribed during latent infection binds to and inhibits *in vitro* binding of infected cell protein 4 to its cognate DNA site. *Proceedings of the National Academy of Sciences, USA* **94**, 10379–10384.

Rice, S. A., Long, M. C., Lam, V., Schaffer, P. A. & Spencer, C. A. (1995). Herpes simplex virus immediate-early protein ICP22 is required for viral modification of host RNA polymerase II and establishment of the normal viral transcription program. *Journal of Virology* **69**, 5550–5559.

Rock, D. L. & Fraser, N. W. (1983). Detection of HSV-1 genome in central nervous system of latently infected mice. *Nature* **302**, 523–525.

Rock, D. L. & Fraser, N. W. (1985). Latent herpes simplex virus type 1 DNA contains two copies of the virion DNA joint region. *Journal of Virology* **55**, 849–852.

Rodahl, E. & Haarr, L. (1997). Analysis of the 2-kilobase latency-associated transcript expressed in PC12 cells productively infected with herpes simplex virus type 1: evidence for a stable, nonlinear structure. *Journal of Virology* **71**, 1703–1707.

Roizman, B. & Sears, A. E. (1987). An inquiry into the mechanisms of herpes simplex virus latency. *Annual Review of Microbiology* **41**, 543–571.

Ryan, A. K. & Rosenfeld, M. G. (1997). POU domain family values: flexibility, partnerships and developmental codes. *Genes & Development* **11**, 1207–1225.

Sacks, W. R. & Schaffer, P. A. (1987). Deletion mutants in the gene encoding the herpes simplex virus type 1 immediate-early protein ICP0 exhibit impaired growth in culture. *Journal of Virology* **61**, 829–839.

Samaniego, L. A., Webb, A. L. & DeLuca, N. A. (1995). Functional interactions between herpes simplex virus immediate-early proteins during infection: gene expression as a consequence of ICP27 and different domains of ICP4. *Journal of Virology* **69**, 5705–5715.

Samaniego, L. A., Wu, N. & DeLuca, N. A. (1997). The herpes simplex virus immediate-early protein ICP0 affects transcription from the viral genome and infected-cell survival in the absence of ICP4 and ICP27. *Journal of Virology* **71**, 4614–4625.

Samaniego, L. A., Neiderhiser, L. & DeLuca, N. A. (1998). Persistence and expression of the herpes simplex virus genome in the absence of immediate-early proteins. *Journal of Virology* **72**, 3307–3320.

Sawtell, N. M. (1997). Comprehensive quantification of herpes simplex virus latency at the single-cell level. *Journal of Virology* **71**, 5423–5431.

Sawtell, N. M. (1998). The probability of *in vivo* reactivation of herpes simplex virus type 1 latency increases with the number of latently infected neurons in the ganglia. *Journal of Virology* **72**, 6888–6892.

Sawtell, N. M. & Thompson, R. L. (1992*a*). Herpes simplex virus type 1 latency-associated transcription unit promotes anatomical site-dependent establishment and reactivation from latency. *Journal of Virology* **66**, 2157–2169.

Sawtell, N. M. & Thompson, R. L. (1992*b*). Rapid *in vivo* reactivation of herpes simplex virus in latently infected murine ganglionic neurons after transient hyperthermia. *Journal of Virology* **66**, 2150–2156.

Sawtell, N. M., Poon, D. K., Tansky, C. S. & Thompson, R. L. (1998). The latent herpes simplex virus type 1 genome copy number in individual neurons is virus strain specific and correlates with reactivation. *Journal of Virology* **72**, 5343–5350.

Sears, A. E., Halliburton, I. W., Meignier, B., Silver, S. & Roizman, B. (1985). Herpes simplex virus 1 mutant deleted in the α22 gene: growth and gene expression in permissive and restrictive cells and establishment of latency in mice. *Journal of Virology* **55**, 338–346.

Sears, A. E., Hukkanen, V., Labow, M. A., Levine, A. J. & Roizman, B. (1991). Expression of herpes simplex virus 1 α transinducing factor (VP16) does not induce reactivation of latent virus or prevent the establishment of latency in mice. *Journal of Virology* **65**, 2929–2935.

Sedarati, F., Margolis, T. P. & Stevens, J. G. (1993). Latent infection can be established with drastically restricted transcription and replication of the HSV-1 genome. *Virology* **192**, 687–691.

Seeler, J. S., Marchio, A., Sitterlin, D., Transby, C. & Dejean, A. (1998). Interaction of sp100 with HP1 proteins: a link between the promyelocytic leukaemia-associated nuclear bodies and the chromatin compartment. *Proceedings of the National Academy of Sciences, USA* **95**, 7316–7321.

Shimeld, C., Hill, T. J., Blyth, W. A. & Easty, D. L. (1990). Reactivation of latent infection and induction of recurrent herpetic eye disease in mice. *Journal of General Virology* **71**, 397–404.

Simmons, A., Slobedman, B., Speck, P., Arthur, J. & Efstathiou, S. (1992). Two patterns of persistence of herpes simplex virus DNA sequences in the nervous systems of latently infected mice. *Journal of General Virology* **73**, 1287–1291.

Slobedman, B., Efstathiou, S. & Simmons, A. (1994). Quantitative analysis of herpes simplex virus DNA in ganglia of mice latently infected with wild-type and thymidine kinase-deficient viral strains. *Journal of General Virology* **75**, 2469–2474.

Smith, R. L., Pizer, L. I., Johnson, E. M. & Wilcox, C. L. (1992). Activation of second messenger pathways reactivates latent herpes simplex virus in neuronal cultures. *Virology* **188**, 311–318.

Smith, R. L., Escudero, J. M. & Wilcox, C. L. (1994). Regulation of the herpes simplex virus latency-associated transcripts during establishment of latency in sensory neurons *in vitro*. *Virology* **202**, 49–60.

Speck, P. G. & Simmons, A. (1991). Divergent molecular pathways of productive and latent infection with a virulent strain of herpes simplex virus type 1. *Journal of Virology* **65**, 4001–4005.

Speck, P. G. & Simmons, A. (1992). Synchronous appearance of antigen-positive and latently infected neurons in spinal ganglia of mice infected with a virulent strain of herpes simplex virus. *Journal of General Virology* **73**, 1281–1285.

Stanberry, L. R., Kern, E. R., Richards, J. T., Abbott, T. M. & Overall, J. C. (1982). Genital herpes in guinea pigs: pathogenesis or the primary infection and description of recurrent disease. *Journal of Infectious Diseases* **146**, 397–404.

Steiner, I., Spivack, J. G., Deshmane, S. L., Ace, C. I., Preston, C. M. & Fraser, N. W. (1990). A herpes simplex virus type 1 mutant containing a non-transinducing Vmw65 protein establishes latent infection *in vivo* in the absence of viral replication and reactivates efficiently from explanted trigeminal ganglia. *Journal of Virology* **64**, 1630–1638.

Stevens, J. G. (1989). Human herpesviruses: a consideration of the latent state. *Microbiological Reviews* **53**, 318–332.

Stevens, J. G. & Cook, M. L. (1971). Latent herpes simplex virus in spinal ganglia of mice. *Science* **173**, 843–845.

Stevens, J. G., Wagner, E. K., Devi-Rao, G. B., Cook, M. L. & Feldman, L. (1987). RNA complementary to a herpesvirus alpha gene mRNA is predominant in latently infected neurons. *Science* **235**, 1056–1059.

Stow, N. D. & Stow, E. C. (1986). Isolation and characterization of a herpes simplex virus type 1 mutant containing a deletion within the gene encoding the immediate early polypeptide Vmw110. *Journal of General Virology* **67**, 2571–2585.

Stow, E. C. & Stow, N. D. (1989). Complementation of a herpes simplex virus type 1 Vmw110 mutant by human cytomegalovirus. *Journal of General Virology* **70**, 695–704.

Sturm, R. A., Das, G. & Herr, W. (1988). The ubiquitous octamer-binding protein Oct-1 contains a POU domain with a homeo box subdomain. *Genes & Development* **2**, 1582–1599.

Suzuki, N., Peter, W., Ciesiolka, T., Gruss, P. & Scholer, H. R. (1993). Mouse Oct-1 contains a composite homeodomain of human Oct-1 and Oct-2. *Nucleic Acids Research* **21**, 245–252.

Tal-Singer, R., Lasner, T. M., Podrzucki, W., Skokotas, A., Leary, J., Berger, J. J. & Fraser, N. W. (1997). Gene expression during reactivation of herpes simplex virus type 1 from latency in the peripheral nervous system is different from that during lytic infection of tissue cultures. *Journal of Virology* **71**, 5268–5276.

Tenser, R. B., Hay, K. A. & Edris, W. A. (1989). Latency-associated transcript but not reactivatable virus is present in sensory ganglion neurons after inoculation of thymidine kinase-negative mutants of herpes simplex virus type 1. *Journal of Virology* **63**, 2861–2865.

Thompson, R. L. & Sawtell, N. M. (1997). The herpes simplex virus type 1 latency-associated transcript gene regulates the establishment of latency. *Journal of Virology* **71**, 5432–5440.

Turner, E. E., Fedtsova, N. & Rosenfeld, M. G. (1996). POU-domain factor expression in the trigeminal ganglion and implications for herpes virus regulation. *NeuroReport* **7**, 2829–2832.

Valyi-Nagy, T., Deshmane, S., Dillner, A. & Fraser, N. W. (1991*a*). Induction of cellular transcription factors in trigeminal ganglia of mice by corneal scarification, herpes simplex virus type 1 infection, and explantation of trigeminal ganglia. *Journal of Virology* **65**, 4142–4152.

Valyi-Nagy, T., Deshmane, S. L., Spivack, J. G., Steiner, I., Ace, C. I., Preston, C. M. & Fraser, N. W. (1991*b*). Investigation of herpes simplex virus type 1 (HSV-1) gene expression and DNA synthesis during the establishment of latent infection by an HSV-1 mutant, *in*1814, that does not replicate in mouse trigeminal ganglia. *Journal of General Virology* **72**, 641–649.

Valyi-Nagy, T., Deshmane, S. L., Raengsakulrach, B., Nicosia, M., Gesser, R. M., Wysocka, M., Dillner, A. & Fraser, N. W. (1992). Herpes simples virus type 1 mutant strain *in*1814 establishes a unique, slowly progressing infection in SCID mice. *Journal of Virology* **66**, 7336–7345.

Wagner, E. K & Bloom, D. C. (1997). Experimental investigation of herpes simplex virus latency. *Clinical Microbiology Reviews* **10**, 419–443.

Wagner, E. K., Guzowski, J. F. & Singh, J. (1995). Transcription of the herpes simplex virus genome during productive and latent infection. *Progress in Nucleic Acid Research and Molecular Biology* **51**, 123–165.

Walz, M. A., Price, R. W. & Notkins, A. L. (1974). Latent ganglionic infection with herpes simplex virus types 1 and 2: viral reactivation *in vivo* after neurectomy. *Science* **184**, 1185–1187.

Watson, R. J. & Clements, J. B. (1980). A herpes simplex virus type 1 function required for early and late virus RNA synthesis. *Nature* **285**, 329–330.

Wheatley, S. C., Kemp, L. M., Wood, J. N. & Latchman, D. S. (1990). Cell lines derived from dorsal root ganglion neurons are nonpermissive for HSV and express only the latency-associated transcript following infection. *Experimental Cell Research* **190**, 243–246.

Wilcox, C. L. & Johnson, E. M. (1987). Nerve growth factor deprivation results in the reactivation of latent herpes simplex virus *in vitro*. *Journal of Virology* **61**, 2311–2315.

Wilcox, C. L. & Johnson, E. M. (1988). Characterization of nerve growth factor-dependent herpes simplex virus latency in neurons *in vitro*. *Journal of Virology* **62**, 393–399.

Wilcox, C. L., Smith, R. L., Freed, C. R. & Johnson, E. M. (1990). Nerve growth factor-dependence of herpes simplex virus latency in peripheral sympathetic and sensory neurons *in vitro*. *Journal of Neuroscience* **104**, 1268–1275.

Wilcox, C. L., Smith, R. L., Everett, R. D. & Mysofski, D. (1997). The herpes simplex virus type 1 immediate-early protein ICP0 is necessary for the efficient establishment of latent infection. *Journal of Virology* **71**, 6777–6785.

Wood, J. N., Lillycrop, K. A., Dent, C. L., Ninkina, N. N., Beech, M. M., Willoughby, J. J., Winter, J. & Latchman, D. S. (1992). Regulation of expression of the neuronal POU protein Oct-2 by nerve growth factor. *Journal of Biological Chemistry* **267**, 17787–17791.

Wu, N., Watkins, S. C., Schaffer, P. A. & DeLuca, N. A. (1996*a*). Prolonged gene expression and cell survival after infection by a herpes simplex virus mutant defective in the immediate-early genes encoding ICP4, ICP27, and ICP22. *Journal of Virology* **70**, 6358–6369.

Wu, T.-T., Su, Y.-H., Block, T. M. & Taylor, J. M. (1996*b*). Evidence that two latency-associated transcripts of herpes simplex virus type 1 are nonlinear. *Journal of Virology* **70**, 5962–5967.

Yao, F. & Schaffer, P. A. (1995). An activity specified by the osteosarcoma line U2OS can substitute functionally for ICP0, a major regulatory protein of herpes simplex virus type 1. *Journal of Virology* **69**, 6249–6258.

Yeh, L. & Schaffer, P. A. (1993). A novel class of transcripts expressed with late kinetics in the absence of ICP4 spans the junction between the long and short segments of the herpes simplex virus type 1 genome. *Journal of Virology* **67**, 7373–7382.

York, I. A., Roop, C., Andrews, D. W., Riddell, S. R., Graham, F. L. & Johnson, D. C. (1994). A cytosolic herpes simplex virus protein inhibits antigen presentation to CD8+ lymphocytes. *Cell* **77**, 525–535.

Zabolotny, J., Krummenacher, C. & Fraser, N. W. (1997). The herpes simplex virus type 1 2·0-kilobase latency-associated transcript is a stable intron which branches at a guanosine. *Journal of Virology* **71**, 4199–4208.